

Seasonal Agriculture Performance Analysis

Major Data Analytics Project

Domain: Agriculture

Project Theme: Seasonal Agriculture Performance

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn

Project Goal

Investigate how agricultural performance varies across seasons and identify meaningful patterns, trends, relationships and differences in the available data.

Important: This is a Data Analytics project. No Machine Learning or dashboard development is required.

1. Problem Statement

Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability and economic conditions. As a result, agricultural performance may differ from one season to another. The objective of this project is to analyze the given agricultural dataset and investigate seasonal differences in agricultural performance by identifying meaningful patterns, trends, relationships and variations within the available data.

Students should independently explore the dataset, formulate relevant analytical questions, select suitable analytical techniques and communicate findings based on evidence from the data.

2. Project Objectives

Understand the structure and quality of the agricultural dataset.

Clean and prepare the data for analysis.

Explore seasonal patterns in the dataset.

Perform descriptive and statistical analysis.

Investigate relationships among relevant variables.

Compare agricultural performance across seasons and other meaningful groups.

Use appropriate univariate, bivariate and multivariate visualizations.

Identify significant findings from the data.

Develop evidence-based insights and recommendations.

✓ 3. Dataset

The dataset contains farm-level agricultural records covering different seasons, locations, crops, environmental conditions, farming practices, production, costs, revenue, profit and resource usage.

Dataset file: seasonal_agriculture_performance_dataset.csv

The notebook is designed so that students can upload/place the CSV in the same folder as this notebook.

```
from ast import increment_lineno
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

%matplotlib inline

sns.set_theme(style="darkgrid")
pd.set_option('display.max_columns', None)
pd.set_option("display.float_format", lambda x: f"{x:,.2f}")
```

```
import pandas as pd

file_path = "/content/seasonal_agriculture_performance_dataset.csv"
df = pd.read_csv(file_path)

print("Dataset loaded successfully...")
```

Dataset loaded successfully...

```
file_path = "/content/seasonal_agriculture_performance_dataset.csv"
df = pd.read_csv(file_path)

print("Dataset loaded successfully...")
```

Dataset loaded successfully...

```
file_path = "/content/seasonal_agriculture_performance_dataset.csv"  
df = pd.read_csv(file_path)
```

```
print("Dataset loaded successfully...")
```

```
Dataset loaded successfully...
```

✓ 4. Initial Data Understanding

```
print("Number of rows:", df.shape[0])  
print("Number of columns:", df.shape[1])
```

```
Number of rows: 4000  
Number of columns: 28
```

df.head(5)

	Farm_ID	State	District	Crop	Season	Farm_Area_Hectares	Rainfall_mm	Avg_Temperature_C	Humidity
0	SF10001	Andhra Pradesh	Rajkot	Wheat	Kharif	0.53	486.40	24.30	
1	SF10002	Maharashtra	Nalgonda	Maize	Kharif	6.53	855.40	25.70	
2	SF10003	Telangana	Warangal	Pulses	Rabi	4.86	455.60	23.90	
3	SF10004	Telangana	Indore	Rice	Kharif	4.50	753.20	29.70	
4	SF10005	Karnataka	Ludhiana	Maize	Zaid	4.21	101.60	34.10	

df.tail(5)

	Farm_ID	State	District	Crop	Season	Farm_Area_Hectares	Rainfall_mm	Avg_Temperature_C	Humidity
3995	SF13996	Karnataka	Rajkot	Wheat	Kharif	4.72	865.40	27.30	
3996	SF13997	Karnataka	Krishna	Chilli	Kharif	8.77	947.50	29.20	
3997	SF13998	Andhra Pradesh	Erode	Pulses	Zaid	10.47	80.00	35.40	
3998	SF13999	Punjab	Warangal	Cotton	Kharif	1.73	475.00	28.20	
3999	SF14000	Punjab	Nashik	Wheat	Kharif	3.31	984.30	25.80	

```
df.sample(5, random_state=40)
```

	Farm_ID	State	District	Crop	Season	Farm_Area_Hectares	Rainfall_mm	Avg_Temperature_C	Humi
2936	SF12937	Tamil Nadu	Nashik	Cotton	Zaid	13.56	441.70	27.30	
2606	SF12607	Gujarat	Ludhiana	Maize	Rabi	1.51	295.60	23.10	
2441	SF12442	Tamil Nadu	Nalgonda	Rice	Rabi	9.82	365.70	22.60	
3415	SF13416	Madhya Pradesh	Guntur	Rice	Rabi	6.37	361.50	23.50	
2658	SF12659	Gujarat	Erode	Sugarcane	Rabi	11.52	658.70	21.10	

```
print(df.columns.tolist())
```

['Farm_ID', 'State', 'District', 'Crop', 'Season', 'Farm_Area_Hectares', 'Rainfall_mm', 'Avg_Temperature'

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4000 entries, 0 to 3999
Data columns (total 28 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Farm_ID              4000 non-null   object
1   State                4000 non-null   object
2   District             4000 non-null   object
```

```
3 Crop 4000 non-null object
4 Season 4000 non-null object
5 Farm_Area_Hectares 4000 non-null float64
6 Rainfall_mm 3952 non-null float64
7 Avg_Temperature_C 4000 non-null float64
8 Humidity_pct 4000 non-null float64
9 Sunlight_Hours_Day 4000 non-null float64
10 Soil_pH 4000 non-null float64
11 Soil_Moisture_pct 3960 non-null float64
12 Nitrogen_kg_ha 4000 non-null float64
13 Phosphorus_kg_ha 4000 non-null float64
14 Potassium_kg_ha 4000 non-null float64
15 Irrigation_Method 4000 non-null object
16 Fertilizer_kg_ha 4000 non-null float64
17 Pesticide_Litre_ha 4000 non-null float64
18 Seed_Quality_Score 4000 non-null float64
19 Yield_Tonnes_Ha 3968 non-null float64
20 Production_Tonnes 4000 non-null float64
21 Market_Price_INR_Tonne 4000 non-null int64
22 Total_Cost_INR 4000 non-null int64
23 Revenue_INR 4000 non-null int64
24 Profit_INR 4000 non-null int64
25 Water_Used_m3 4000 non-null int64
26 Water_Efficiency_t_per_1000m3 4000 non-null float64
27 Disease_Pest_Risk_pct 4000 non-null float64
dtypes: float64(17), int64(5), object(6)
memory usage: 875.1+ KB
```


df.describe()

	Farm_Area_Hectares	Rainfall_mm	Avg_Temperature_C	Humidity_pct	Sunlight_Hours_Day	Soil_pH	Soil_
count	4,000.00	3,952.00	4,000.00	4,000.00	4,000.00	4,000.00	
mean	7.89	601.15	26.82	63.21	7.32	6.70	
std	4.22	288.93	3.82	11.96	1.12	0.64	
min	0.50	80.00	16.20	25.00	3.50	5.20	
25%	4.26	370.20	23.90	54.77	6.60	6.26	
50%	7.89	582.00	26.90	63.00	7.30	6.69	
75%	11.65	834.95	29.60	71.90	8.10	7.14	
max	15.00	1,395.20	39.70	95.00	11.00	8.40	

df.describe(include="object")

	Farm_ID	State	District	Crop	Season	Irrigation_Method
count	4000	4000	4000	4000	4000	4000
unique	4000	8	10	8	3	4
top	SF13984	Andhra Pradesh	Ludhiana	Rice	Kharif	Flood
freq	1	529	427	690	1779	1310

✓ 5 Data Quality Analysis

Before performing analysis, investigate missing values, duplicates, data types and unusual values.

Student task: Explain what you observe and justify each cleaning decision.

```
missing = df.isnull().sum().sort_values(ascending=False)
missing_percentage = (df.isnull().mean()*100).sort_values(ascending=False)

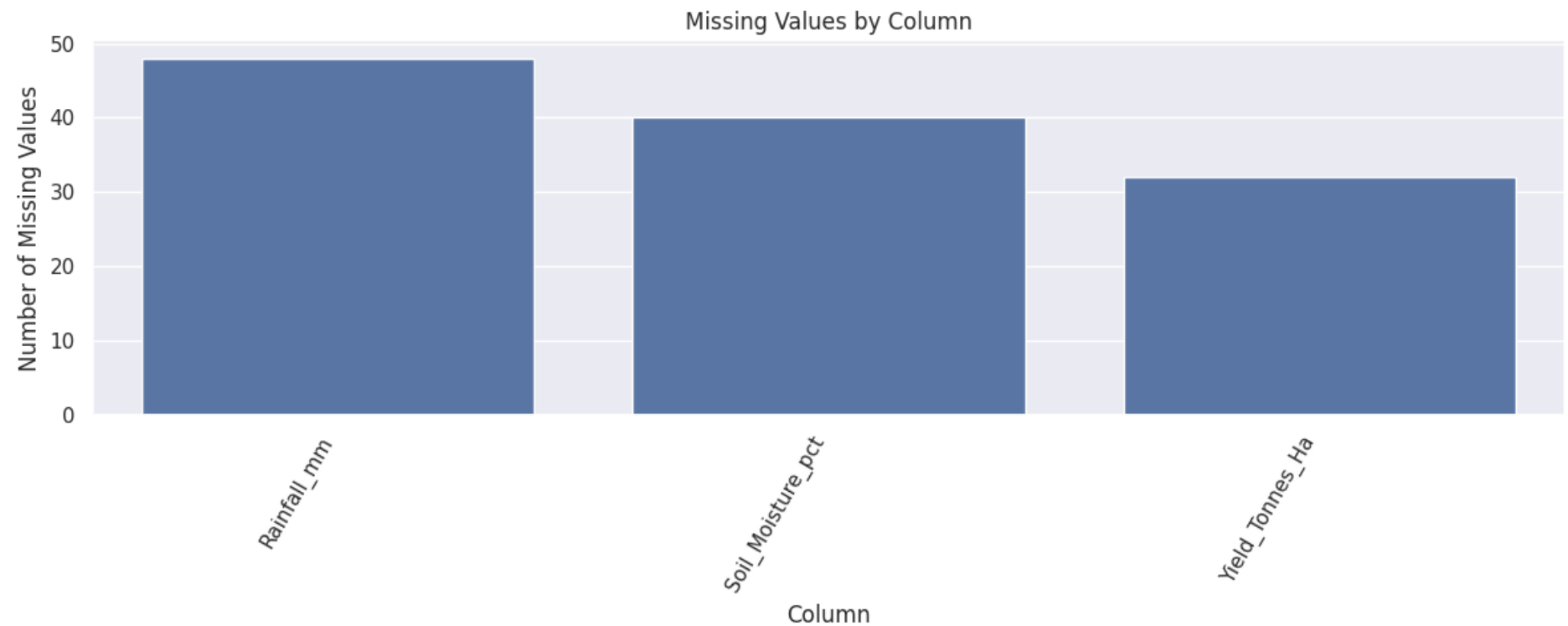
missing_summary = pd.DataFrame({
    "Missing_Count": missing,
    "Missing_Percentage": missing_percentage.round(2)
})

missing_summary[missing_summary["Missing_Count"]>0]
```

	Missing_Count	Missing_Percentage
Rainfall_mm	48	1.20
Soil_Moisture_pct	40	1.00
Yield_Tonnes_Ha	32	0.80

```
plt.figure(figsize=(12, 5))
missing_plot = missing[missing > 0]
if len(missing_plot) > 0:
    sns.barplot(x=missing_plot.index, y=missing_plot.values)
    plt.xticks(rotation=60, ha="right")
```

```
plt.ylabel("Number of Missing Values")
plt.xlabel("Column")
plt.title("Missing Values by Column")
plt.tight_layout()
plt.show()
else:
    print("No missing values found.")
```



Double-click (or enter) to edit

```
duplicate_count = df.duplicated().sum()  
print("Number of duplication rows: ", duplicate_count)
```

Number of duplication rows: 0

```
df = df.drop_duplicates().reset_index(drop=True)  
  
print("Shapes after removing duplicates:",df.shape)
```

Shapes after removing duplicates: (4000, 28)

df.dtypes

	0
Farm_ID	object
State	object
District	object
Crop	object
Season	object
Farm_Area_Hectares	float64
Rainfall mm	float64

Avg_Temperature_C	float64
Humidity_pct	float64
Sunlight_Hours_Day	float64
Soil_pH	float64
Soil_Moisture_pct	float64
Nitrogen_kg_ha	float64
Phosphorus_kg_ha	float64
Potassium_kg_ha	float64
Irrigation_Method	object
Fertilizer_kg_ha	float64
Pesticide_Litre_ha	float64
Seed_Quality_Score	float64
Yield_Tonnes_Ha	float64
Production_Tonnes	float64
Market_Price_INR_Tonne	int64
Total_Cost_INR	int64
Revenue_INR	int64
Profit_INR	int64
Water_Used_m3	int64

Water_Efficiency_t_per_1000m3 float64

Disease_Pest_Risk_pct float64

dtype: object

```
categorical_columns = df.select_dtypes(include="object").columns
```

```
for col in categorical_columns:
    print(f"\n{col}:")
    print(df[col].nunique(), "unique values")
    print(df[col].dropna().unique()[:20])
```

Farm_ID:

4000 unique values

```
['SF10001' 'SF10002' 'SF10003' 'SF10004' 'SF10005' 'SF10006' 'SF10007'
 'SF10008' 'SF10009' 'SF10010' 'SF10011' 'SF10012' 'SF10013' 'SF10014'
 'SF10015' 'SF10016' 'SF10017' 'SF10018' 'SF10019' 'SF10020']
```

State:

8 unique values

```
['Andhra Pradesh' 'Maharashtra' 'Telangana' 'Karnataka' 'Gujarat'
 'Tamil Nadu' 'Punjab' 'Madhya Pradesh']
```

District:

10 unique values

```
['Rajkot' 'Nalgonda' 'Warangal' 'Indore' 'Ludhiana' 'Raichur' 'Guntur'
 'Krishna' 'Nashik' 'Erode']
```

Crop:

8 unique values

```
['Wheat' 'Maize' 'Pulses' 'Rice' 'Cotton' 'Chilli' 'Groundnut' 'Sugarcane']
```

```
Season:
3 unique values
['Kharif' 'Rabi' 'Zaid']

Irrigation_Method:
4 unique values
['Drip' 'Flood' 'Rainfed' 'Sprinkler']
```

```
df.isnull().sum().sort_values(ascending=False).head(10)
```

	0
Rainfall_mm	48
Soil_Moisture_pct	40
Yield_Tonnes_Ha	32
Farm_ID	0
Crop	0
District	0
Farm_Area_Hectares	0
Season	0
Avg_Temperature_C	0
Humidity_pct	0

dtype: int64

✓ Missing Value Treatment

Choose an appropriate strategy based on the nature of each variable.

Possible approaches:

- Remove records only when justified.
- Fill numerical values using an appropriate statistic when appropriate.
- Fill categorical values using a suitable category or method.
- Clearly document why the chosen approach is reasonable.

Do not blindly replace every missing value with zero.

```
import numpy as np

numeric_columns = df.select_dtypes(include=np.number).columns

for col in numeric_columns:
    if df[col].isnull().sum()>0:
        df[col] = df[col].fillna(df[col].median())

print("Remaining missing values:", df.isnull().sum().sum())

Remaining missing values: 0
```


✓ 6. Feature / Variable Review

Review the available variables and classify them conceptually as:

- Identifier variables
- Categorical variables
- Numerical variables
- Environmental variables
- Operational variables
- Production/performance variables
- Economic variables

Student task: Explain which variables are most relevant to the seasonal-performance problem and why.

```

numeric_columns = df.select_dtypes(include=np.number).columns.tolist()
categorical_columns = df.select_dtypes(include="object").columns.tolist()

print("Numerical columns:")
print(numeric_columns)

print("\ncategorical columns:")
print(categorical_columns)

```

```

Numerical columns:
['Farm_Area_Hectares', 'Rainfall_mm', 'Avg_Temperature_C', 'Humidity_pct', 'Sunlight_Hours_Day', 'Soil_ph']

categorical columns:
['Farm_ID', 'State', 'District', 'Crop', 'Season', 'Irrigation_Method']

```

✓ 7. Statistical Analysis

Perform descriptive statistical analysis to understand the distribution and variation of the data.

Students should interpret the statistics rather than simply displaying them.

```

stats = df[numeric_columns].describe().T
stats["range"] = stats["max"] - stats["min"]
stats["IQR"] = stats["75%"] - stats["25%"]

```

```
stats
```

	count	mean	std	min	25%	50%	75%	max
Farm_Area_Hectares	100	1.2	0.8	0.1	0.5	1.0	1.5	2.5
Rainfall_mm	100	1200	400	500	800	1100	1300	1800
Avg_Temperature_C	100	28.5	2.5	25.0	27.0	29.0	30.0	32.0
Humidity_pct	100	75.0	10.0	60.0	65.0	75.0	85.0	95.0
Sunlight_Hours_Day	100	6.5	1.5	4.0	5.0	6.0	7.0	8.0
Soil_ph	100	6.5	0.5	5.5	6.0	6.5	7.0	7.5

Farm_Area_Hectares	4000.0	7.888265	4.223764	0.500	4.260	7.8950	11.6500
Rainfall_mm	3952.0	601.152657	288.925025	80.000	370.200	582.0000	834.9500
Avg_Temperature_C	4000.0	26.818900	3.818180	16.200	23.900	26.9000	29.6000
Humidity_pct	4000.0	63.210500	11.960834	25.000	54.775	63.0000	71.9000
Sunlight_Hours_Day	4000.0	7.323050	1.121186	3.500	6.600	7.3000	8.1000
Soil_pH	4000.0	6.701010	0.641461	5.200	6.260	6.6900	7.1400
Soil_Moisture_pct	3960.0	26.508232	6.712484	8.000	21.700	26.4000	31.3000
Nitrogen_kg_ha	4000.0	120.250825	31.216452	40.000	98.500	120.1000	141.2250
Phosphorus_kg_ha	4000.0	57.781850	17.015074	15.000	46.200	57.5000	69.4000
Potassium_kg_ha	4000.0	104.760050	27.742681	30.000	86.200	105.1000	123.6000
Fertilizer_kg_ha	4000.0	186.036000	60.260583	40.000	143.975	186.2500	227.2000
Pesticide_Litre_ha	4000.0	5.070935	1.978879	0.500	3.690	5.1200	6.4000
Seed_Quality_Score	4000.0	0.826138	0.101747	0.650	0.740	0.8300	0.9100
Yield_Tonnes_Ha	3968.0	5.270595	13.361398	0.300	1.050	1.7400	2.8400
Production_Tonnes	4000.0	43.246210	126.880521	0.150	5.020	11.7700	24.3550
Market_Price_INR_Tonne	4000.0	44714.166250	30266.161922	2594.000	21429.250	25913.0000	68279.0000
Total_Cost_INR	4000.0	526303.583000	294419.140911	26826.000	274577.500	519082.5000	761996.5000
Revenue_INR	4000.0	637860.048000	636974.710054	6107.000	207517.250	456830.5000	836874.7500
Profit_INR	4000.0	111556.465000	545081.890509	-1019223.000	-148731.750	3673.0000	216187.7500
Water_Used_m3	4000.0	6045.570250	5572.944306	128.000	2268.750	4435.0000	7829.7500

Water_Efficiency_t_per_1000m3	4000.0	5.385214	8.838101	0.135	1.661	2.9375	5.1502
Disease_Pest_Risk_pct	4000.0	46.365875	12.424417	5.000	37.300	46.4000	54.9000

```
ststistical_summary = pd.DataFrame({
    "mean": df[numeric_columns].mean(),
    "median": df[numeric_columns].median(),
    "std_Dev": df[numeric_columns].std(),
    "min": df[numeric_columns].min(),
    "max": df[numeric_columns].max(),
    "range": df[numeric_columns].max() - df[numeric_columns].min(),
})
```

```
ststistical_summary
```

	mean	median	std_Dev	min	max	range
Farm_Area_Hectares	7.888265	7.8950	4.223764	0.500	15.00	14.500
Rainfall_mm	601.152657	582.0000	288.925025	80.000	1395.20	1315.200
Avg_Temperature_C	26.818900	26.9000	3.818180	16.200	39.70	23.500
Humidity_pct	63.210500	63.0000	11.960834	25.000	95.00	70.000
Sunlight_Hours_Day	7.323050	7.3000	1.121186	3.500	11.00	7.500
Soil_pH	6.701010	6.6900	0.641461	5.200	8.40	3.200
Soil_Moisture_pct	26.508232	26.4000	6.712484	8.000	47.00	39.000
Nitrogen_kg_ha	120.250825	120.1000	31.216452	40.000	220.00	180.000
Phosphorus_kg_ha	57.781850	57.5000	17.015074	15.000	120.00	105.000

Potassium_kg_ha	104.760050	105.1000	27.742681	30.000	200.00	170.000
Fertilizer_kg_ha	186.036000	186.2500	60.260583	40.000	400.00	360.000
Pesticide_Litre_ha	5.070935	5.1200	1.978879	0.500	12.08	11.580
Seed_Quality_Score	0.826138	0.8300	0.101747	0.650	1.00	0.350
Yield_Tonnes_Ha	5.270595	1.7400	13.361398	0.300	101.44	101.140
Production_Tonnes	43.246210	11.7700	126.880521	0.150	1424.40	1424.250
Market_Price_INR_Tonne	44714.166250	25913.0000	30266.161922	2594.000	131240.00	128646.000
Total_Cost_INR	526303.583000	519082.5000	294419.140911	26826.000	1349990.00	1323164.000
Revenue_INR	637860.048000	456830.5000	636974.710054	6107.000	5080900.00	5074793.000
Profit_INR	111556.465000	3673.0000	545081.890509	-1019223.000	4345421.00	5364644.000
Water_Used_m3	6045.570250	4435.0000	5572.944306	128.000	40902.00	40774.000
Water_Efficiency_t_per_1000m3	5.385214	2.9375	8.838101	0.135	79.80	79.665
Disease_Pest_Risk_pct	46.365875	46.4000	12.424417	5.000	88.90	83.900

```
season_summary = df.groupby("Season")[numeric_columns].agg(["mean", "median", "std"])
season_summary
```

	Farm_Area_Hectares			Rainfall_mm			Avg_Temperature_C			Humidity_pct	...
	mean	median	std	mean	median	std	mean	median	std	mean	...
Season											
Kharif	7.944924	8.180	4.282328	852.084205	854.75	182.552740	28.454019	28.5	2.510166	71.812591	...
Rabi	7.752207	7.680	4.179402	436.000124	430.30	175.659664	23.489183	23.5	2.459561	57.893178	...
Zaid	8.091246	8.085	4.162594	299.421784	283.30	153.980457	31.042088	31.0	2.554911	52.012121	...

3 rows x 66 columns

✓ 8. Univariate Analysis

Univariate analysis examines one variable at a time.

Create and interpret suitable plots for:

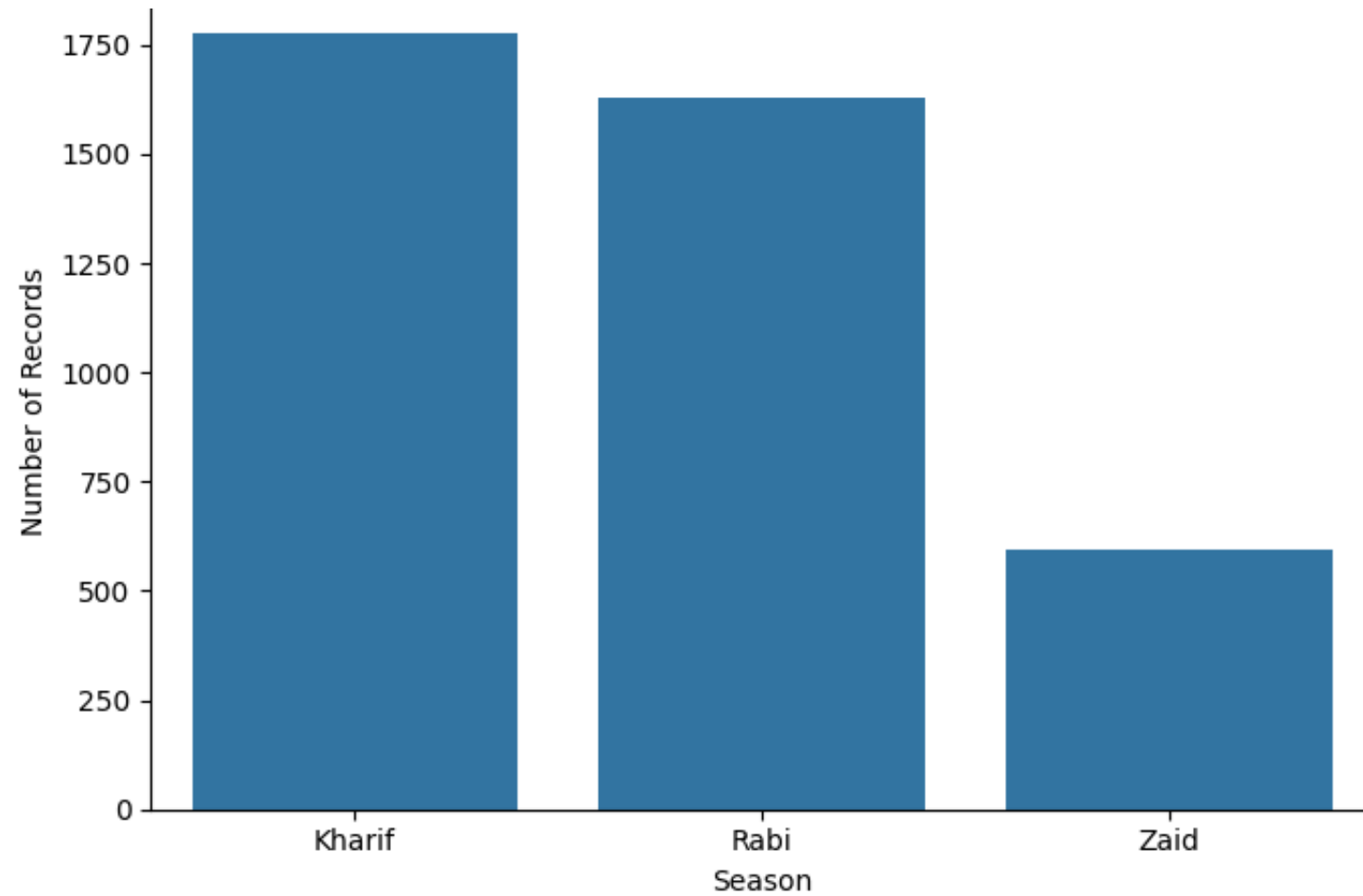
- Important categorical variables
- Important numerical Variables
- Seasonal distribution
- Production/performance distributions
- Economic distributions

The examples below are starting points. Students should add or modify plots based on their own analytical questions.

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(7, 5))
sns.countplot(data=df, x="Season")
plt.title("Distribution of Records by Seasons")
plt.xlabel("Season")
plt.ylabel("Number of Records")
plt.tight_layout()
plt.show()
```

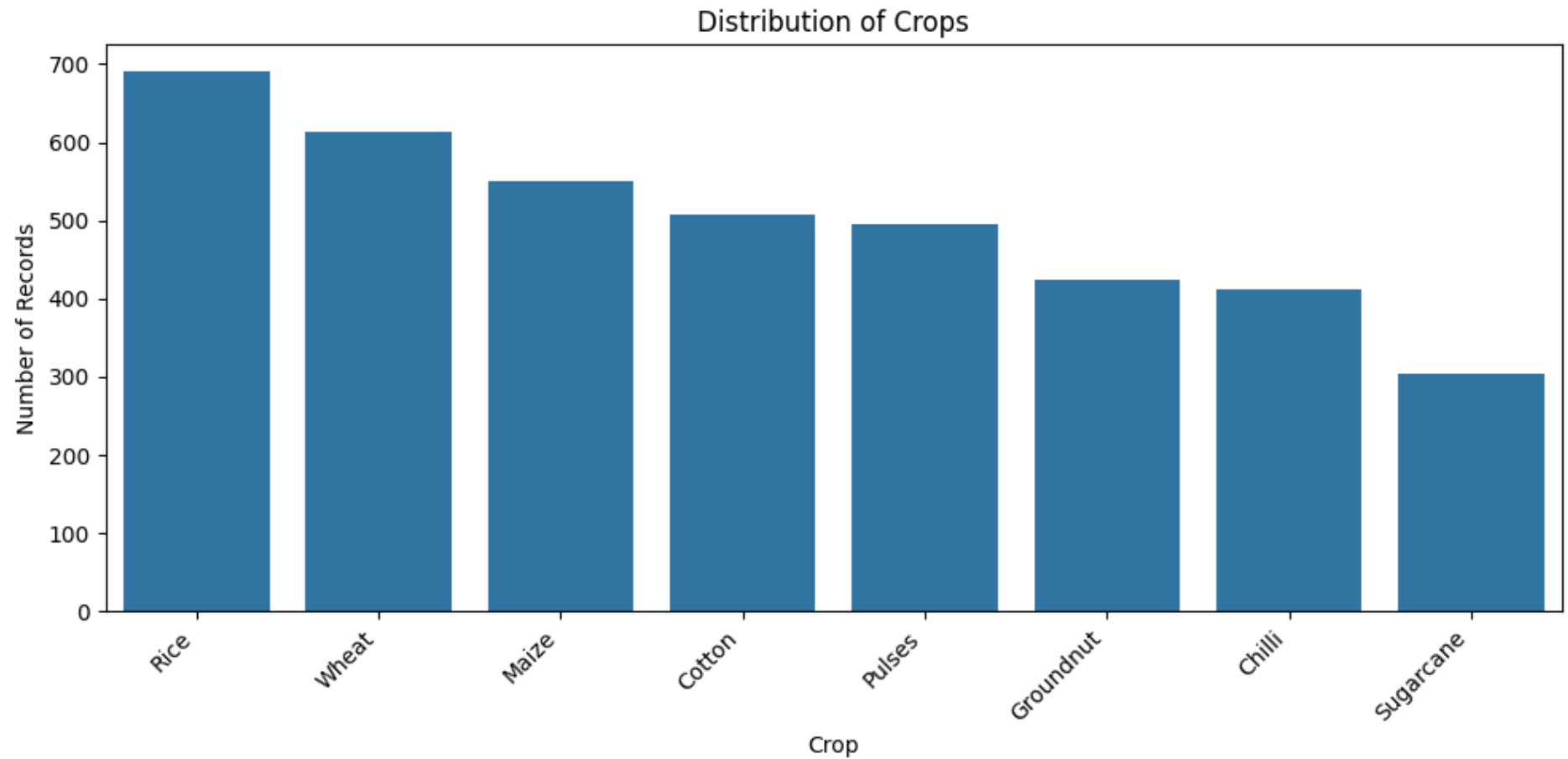
Distribution of Records by Seasons



```
plt.figure(figsize=(10, 5))
sns.countplot(data=df, x="Crop", order=df["Crop"].value_counts().index)
plt.xticks(rotation=45, ha="right")
plt.title("Distribution of Crops")
plt.xlabel("Crop")
plt.ylabel("Number of Records")
```

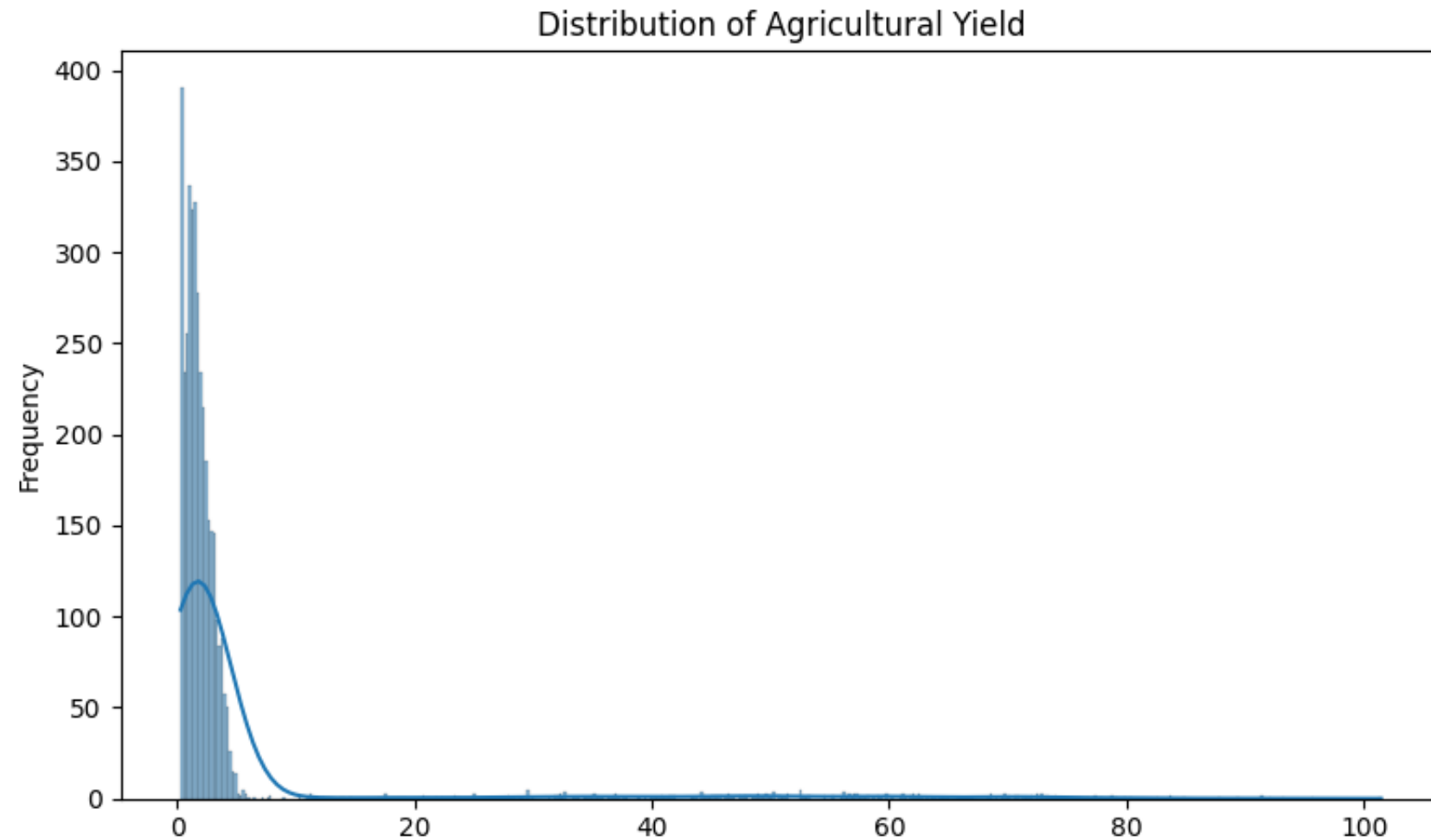


```
plt.tight_layout()  
plt.show()
```



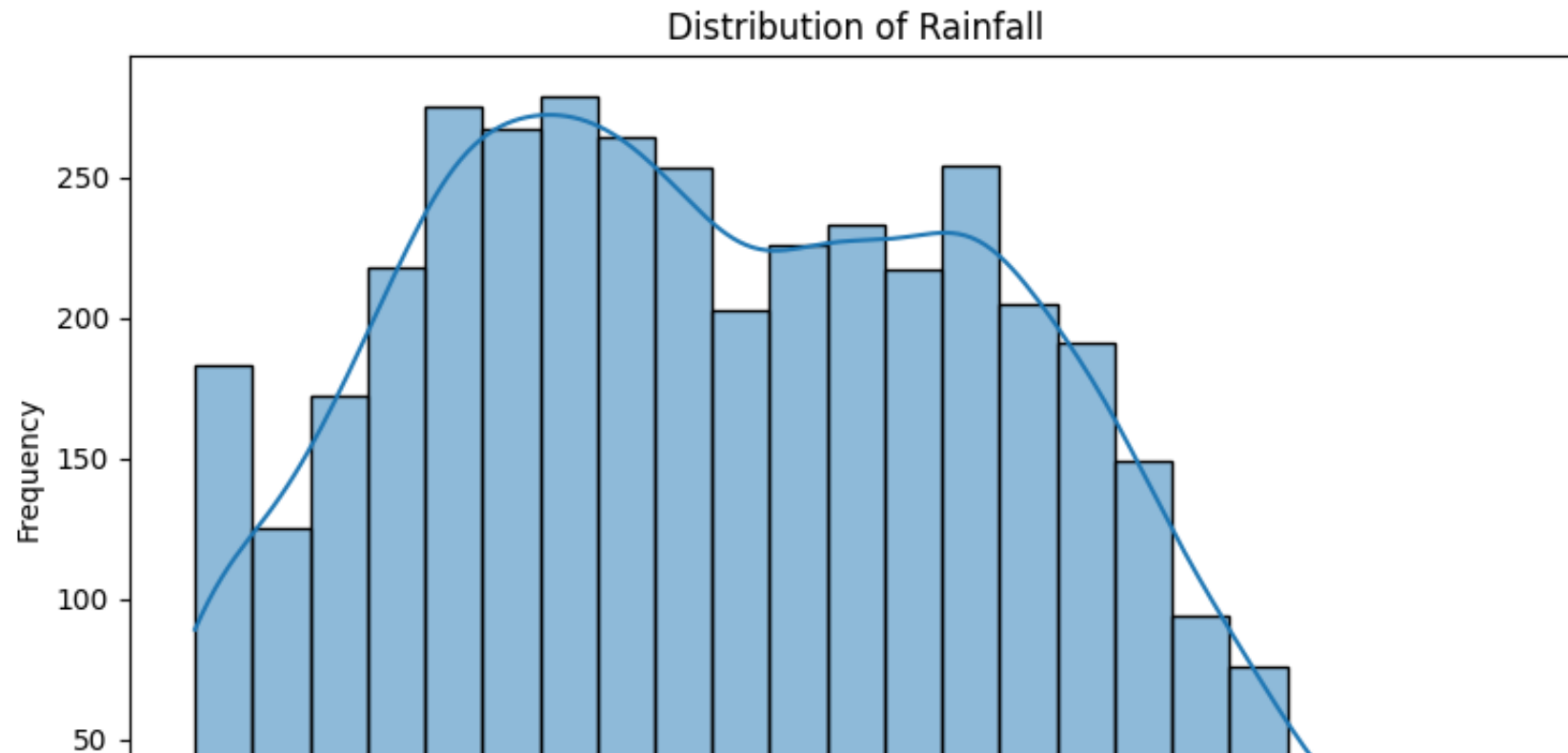
```
plt.figure(figsize=(8, 5))
```

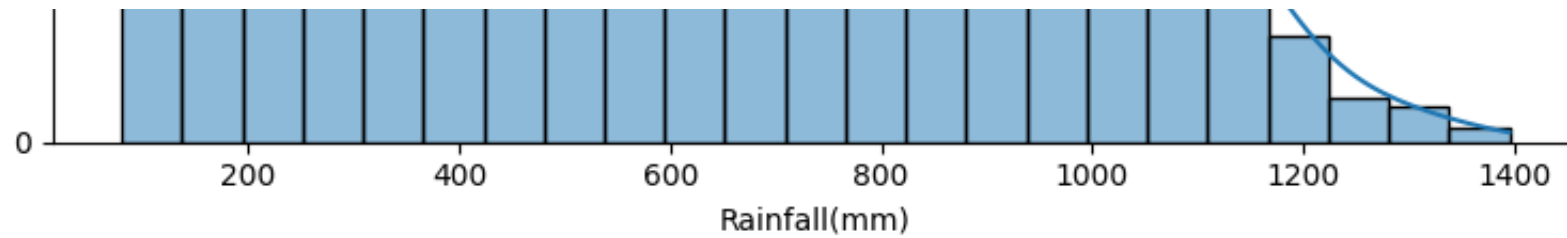
```
sns.histplot(data=df, x="Yield_Tonnes_Ha", kde=True)
plt.title("Distribution of Agricultural Yield")
plt.xlabel("Yield(tonnes/ha)")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
#
```



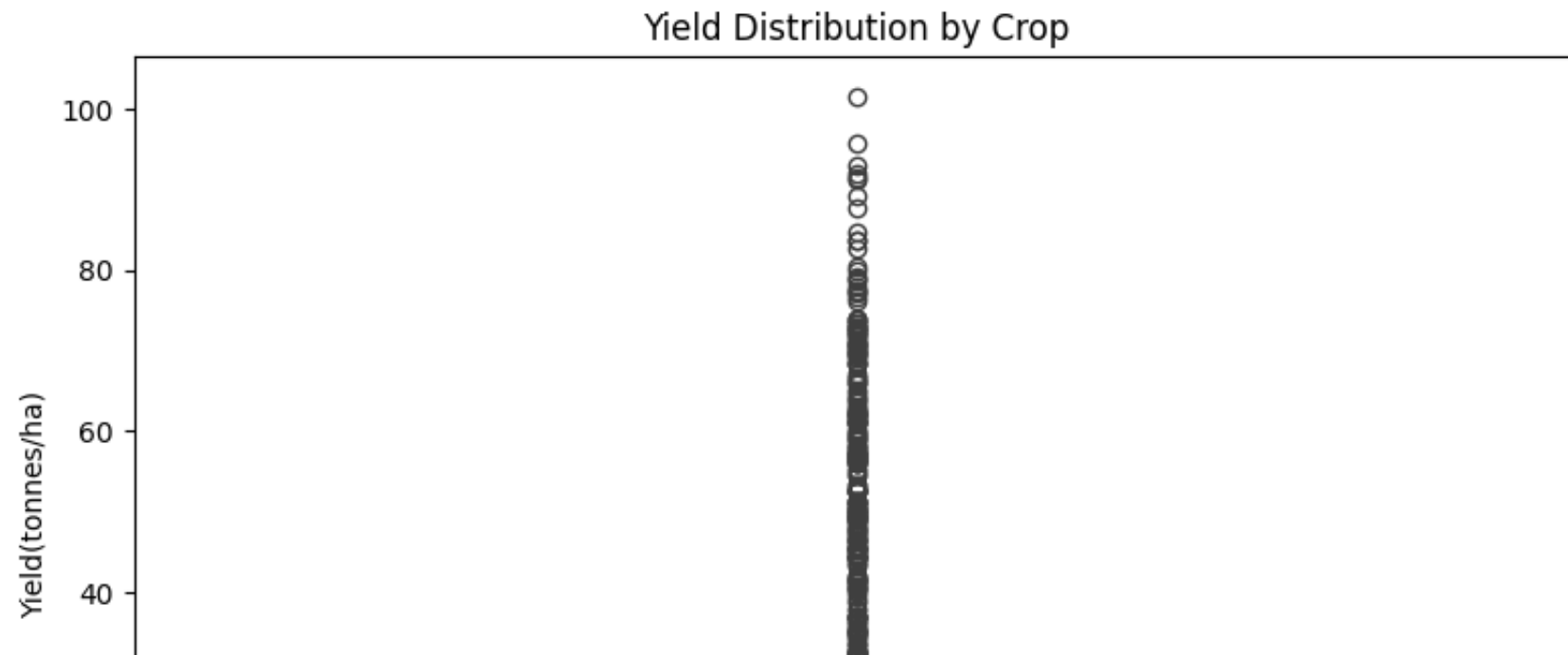
Yield(tonnes/ha)

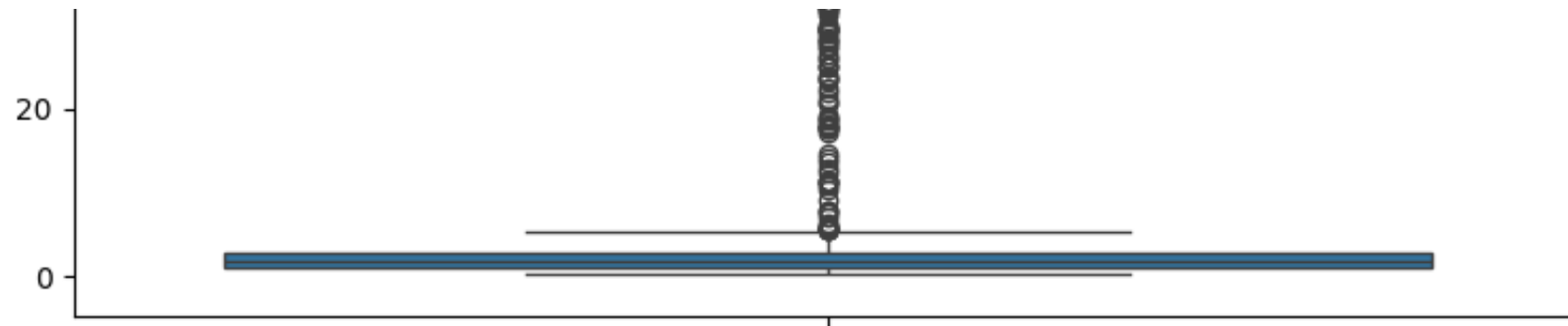
```
plt.figure(figsize=(8, 5))
sns.histplot(data=df, x="Rainfall_mm", kde=True)
plt.title("Distribution of Rainfall")
plt.xlabel("Rainfall(mm)")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
```





```
plt.figure(figsize=(8, 5))
sns.boxplot(data=df, y="Yield_Tonnes_Ha")
plt.title("Yield Distribution by Crop")
plt.ylabel("Yield(tonnes/ha)")
plt.tight_layout()
plt.show()
```





✓ 9. Outlier Analysis

Investigate potential outliers using statistical and visual methods.

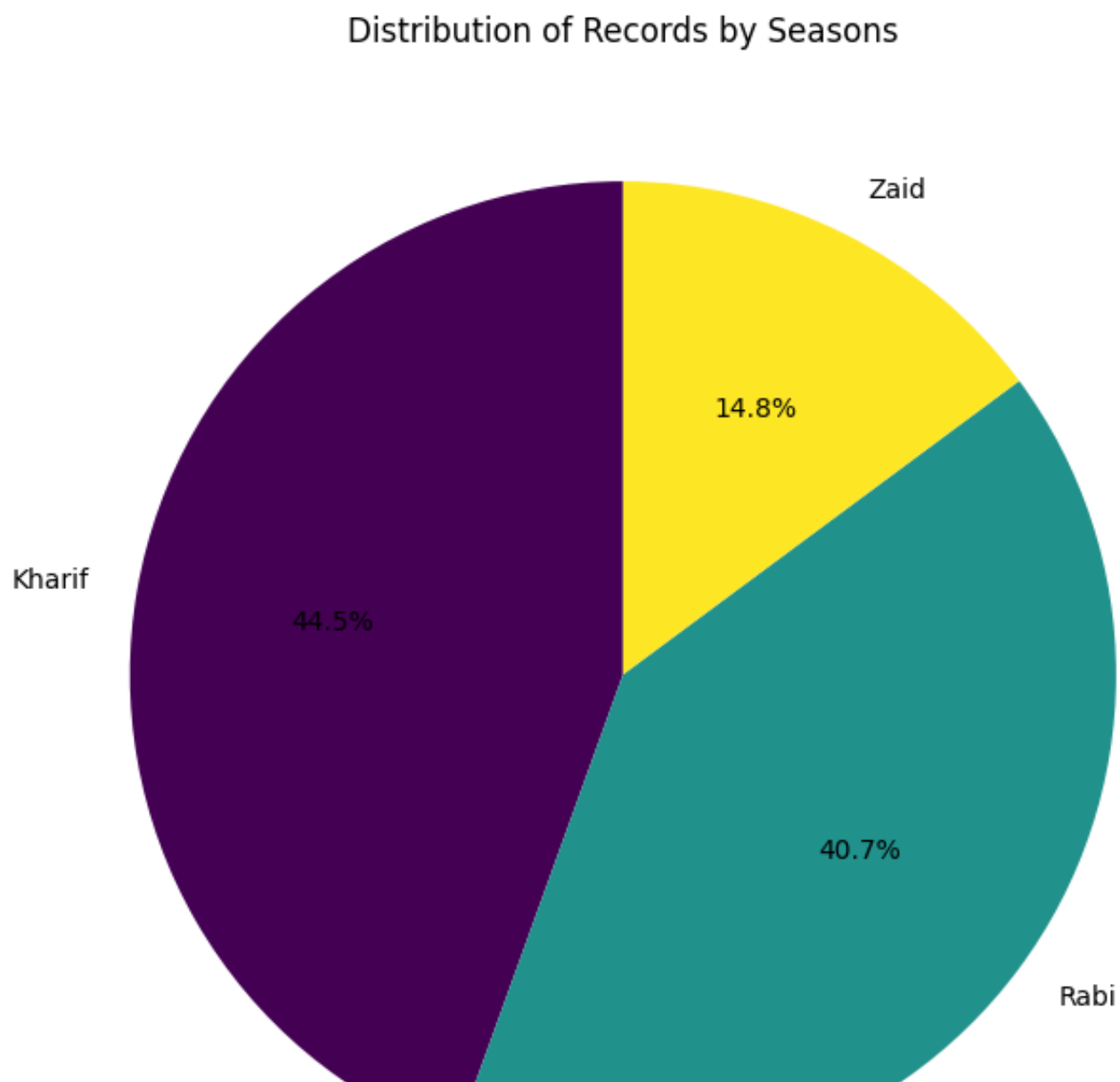
Student task: Do not automatically delete outliers. Decide whether an extreme value is:

- A data-entry problem,
- A legitimate observation,
- Or something requiring further investigation.

Univariate: Season distribution (Pie Chart)

```
plt.figure(figsize=(7, 7))
df['Season'].value_counts().plot.pie(autopct='%1.1f%%', startangle=90, cmap='viridis')
plt.title('Distribution of Records by Seasons')
plt.ylabel('')
plt.tight_layout()
```

```
plt.show()
```





```

outlier_summary = []

for col in numeric_columns:
    q1 = df[col].quantile(0.25)
    q3 = df[col].quantile(0.75)
    iqr = q3 - q1
    lower_bound = q1 - 1.5 * iqr
    upper_bound = q3 + 1.5 * iqr
    count = ((df[col] < lower_bound) | (df[col] > upper_bound)).sum()

    outlier_summary.append({
        "Column": col,
        "Lower_Bound": lower_bound,
        "Upper_Bound": upper_bound,
        "Outlier_Count": count,
        "Outlier_Percentage": round(count/len(df)*100,2)
    })

outlier_summary = pd.DataFrame(outlier_summary).sort_values(by="Outlier_Count", ascending=False)
outlier_summary

```

	Column	Lower_Bound	Upper_Bound	Outlier_Count	Outlier_Percentage
18	Profit_INR	-696111.000000	7.635670e+05	404	10.10

14	Production_Tonnes	-23.982500	5.335750e+01	333	8.33
20	Water_Efficiency_t_per_1000m3	-3.572875	1.038412e+01	322	8.05
13	Yield_Tonnes_Ha	-1.635000	5.525000e+00	302	7.55
19	Water_Used_m3	-6072.750000	1.617125e+04	276	6.90
17	Revenue_INR	-736519.000000	1.780911e+06	240	6.00
9	Potassium_kg_ha	30.100000	1.797000e+02	32	0.80
4	Sunlight_Hours_Day	4.350000	1.035000e+01	26	0.65
8	Phosphorus_kg_ha	11.400000	1.042000e+02	16	0.40
10	Fertilizer_kg_ha	19.137500	3.520375e+02	15	0.38
7	Nitrogen_kg_ha	34.412500	2.053125e+02	10	0.25
21	Disease_Pest_Risk_pct	10.900000	8.130000e+01	10	0.25
11	Pesticide_Litre_ha	-0.375000	1.046500e+01	8	0.20
3	Humidity_pct	29.087500	9.758750e+01	6	0.15
6	Soil_Moisture_pct	7.300000	4.570000e+01	3	0.08
2	Avg_Temperature_C	15.350000	3.815000e+01	2	0.05
0	Farm_Area_Hectares	-6.825000	2.273500e+01	0	0.00
1	Rainfall_mm	-326.925000	1.532075e+03	0	0.00
5	Soil_pH	4.940000	8.460000e+00	0	0.00
12	Seed_Quality_Score	0.485000	1.165000e+00	0	0.00
16	Total_Cost_INR	-456551.000000	1.493125e+06	0	0.00

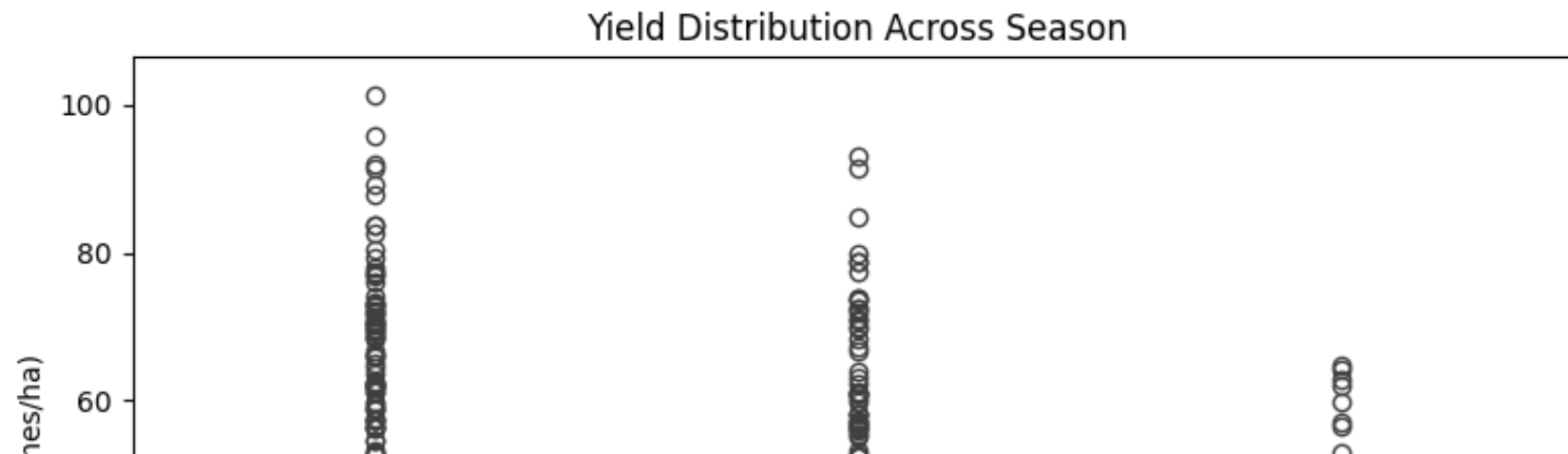
	total_cost_INR	total_cost_USD	total_yield_tonnes	total_yield_usd
15	Market_Price_INR_Tonne	-48845.375000	1.385536e+05	0

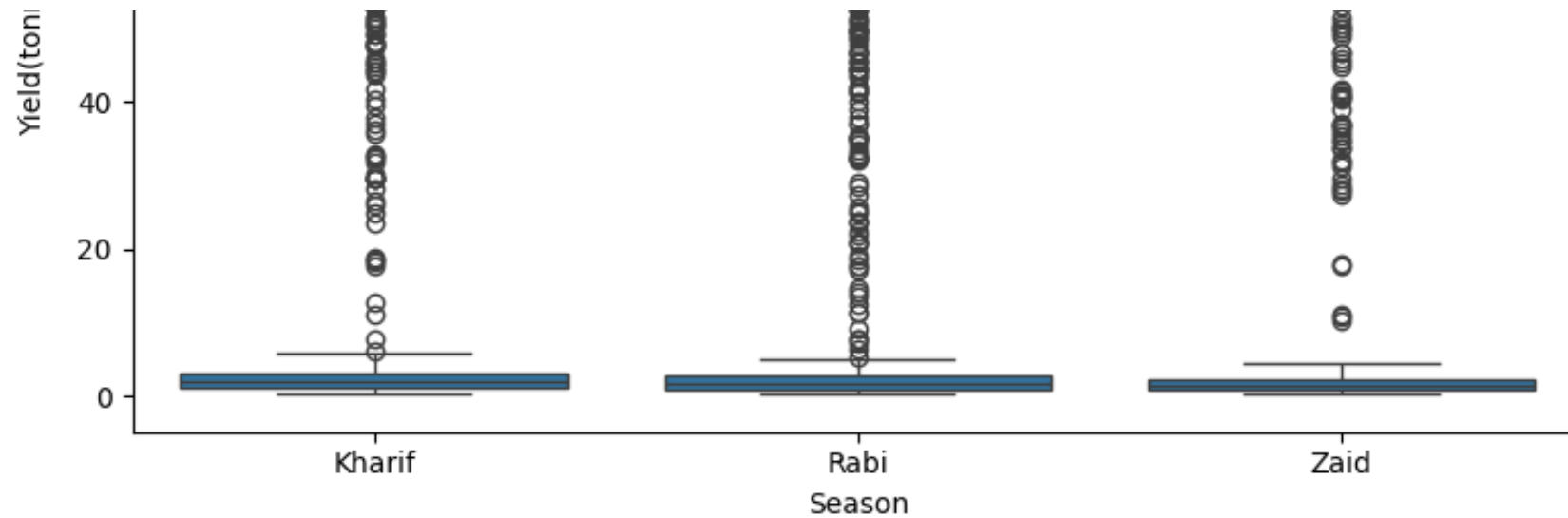
10. Bivariate Analysis

Bivariate analysis examines relationships or differences between two variables.

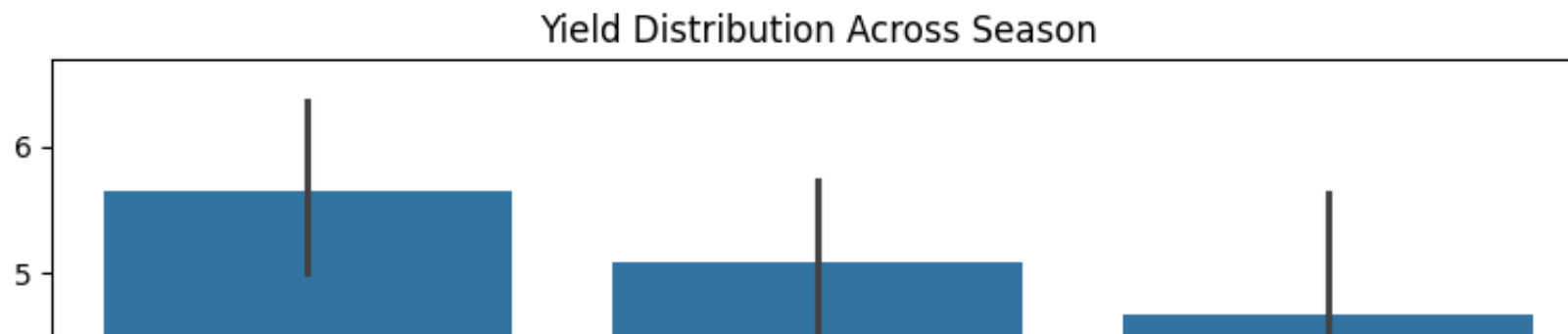
Students should investigate relationships that are relevant to the seasonal agriculture problem.

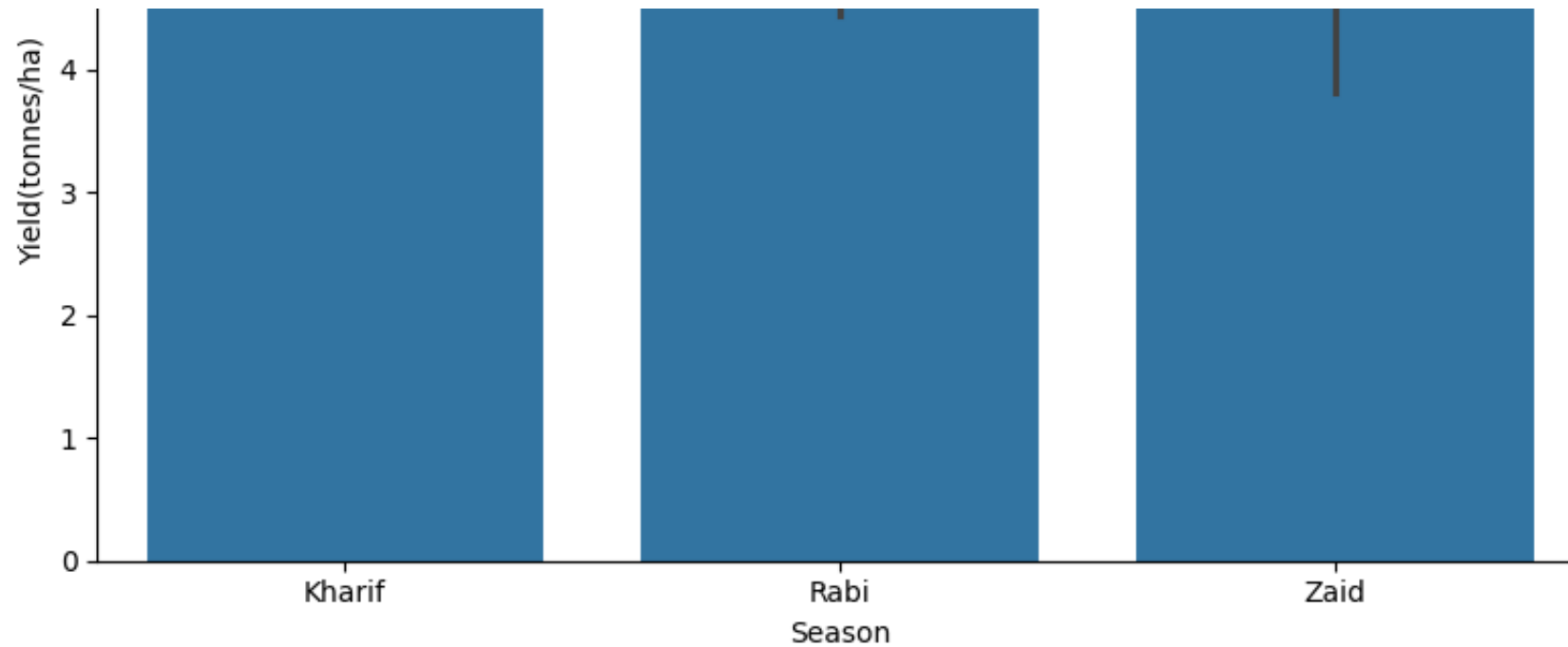
```
plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x="Season", y="Yield_Tonnes_Ha")
plt.title("Yield Distribution Across Season")
plt.xlabel("Season")
plt.ylabel("Yield(tonnes/ha)")
plt.tight_layout()
plt.show()
```





```
plt.figure(figsize=(8, 5))
sns.barplot(data=df, x="Season", y="Yield_Tonnes_Ha")
plt.title("Yield Distribution Across Season")
plt.xlabel("Season")
plt.ylabel("Yield (tonnes/ha)")
plt.tight_layout()
plt.show()
```

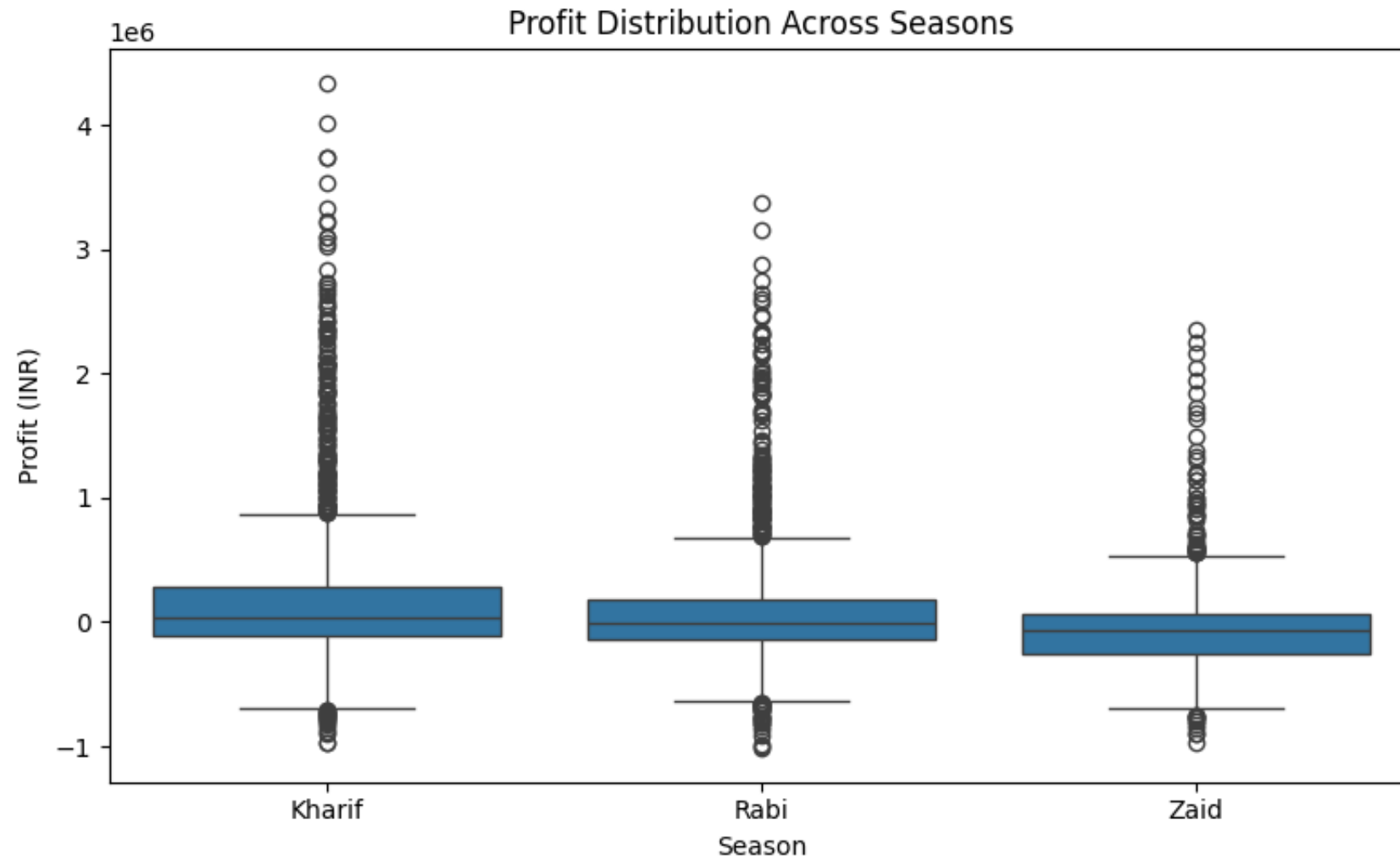




```
average_yield = df['Yield_Tonnes_Ha'].mean()
print(f"Average Yield: {average_yield:.2f} Tonnes/Ha")
```

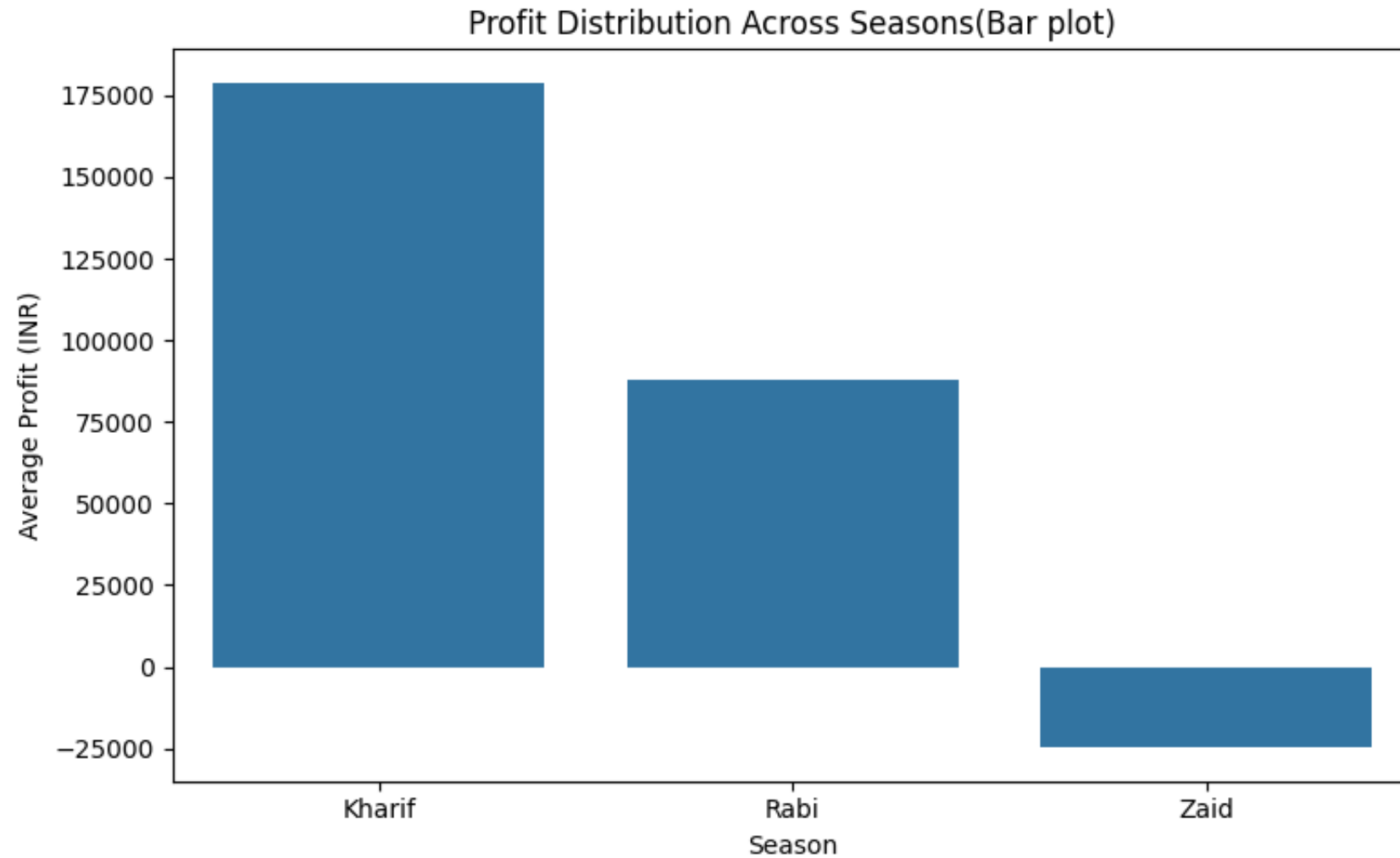
Average Yield: 5.27 Tonnes/Ha

```
plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x="Season", y="Profit_INR")
plt.title("Profit Distribution Across Seasons")
plt.xlabel("Season")
plt.ylabel("Profit (INR)")
plt.tight_layout()
plt.show()
```



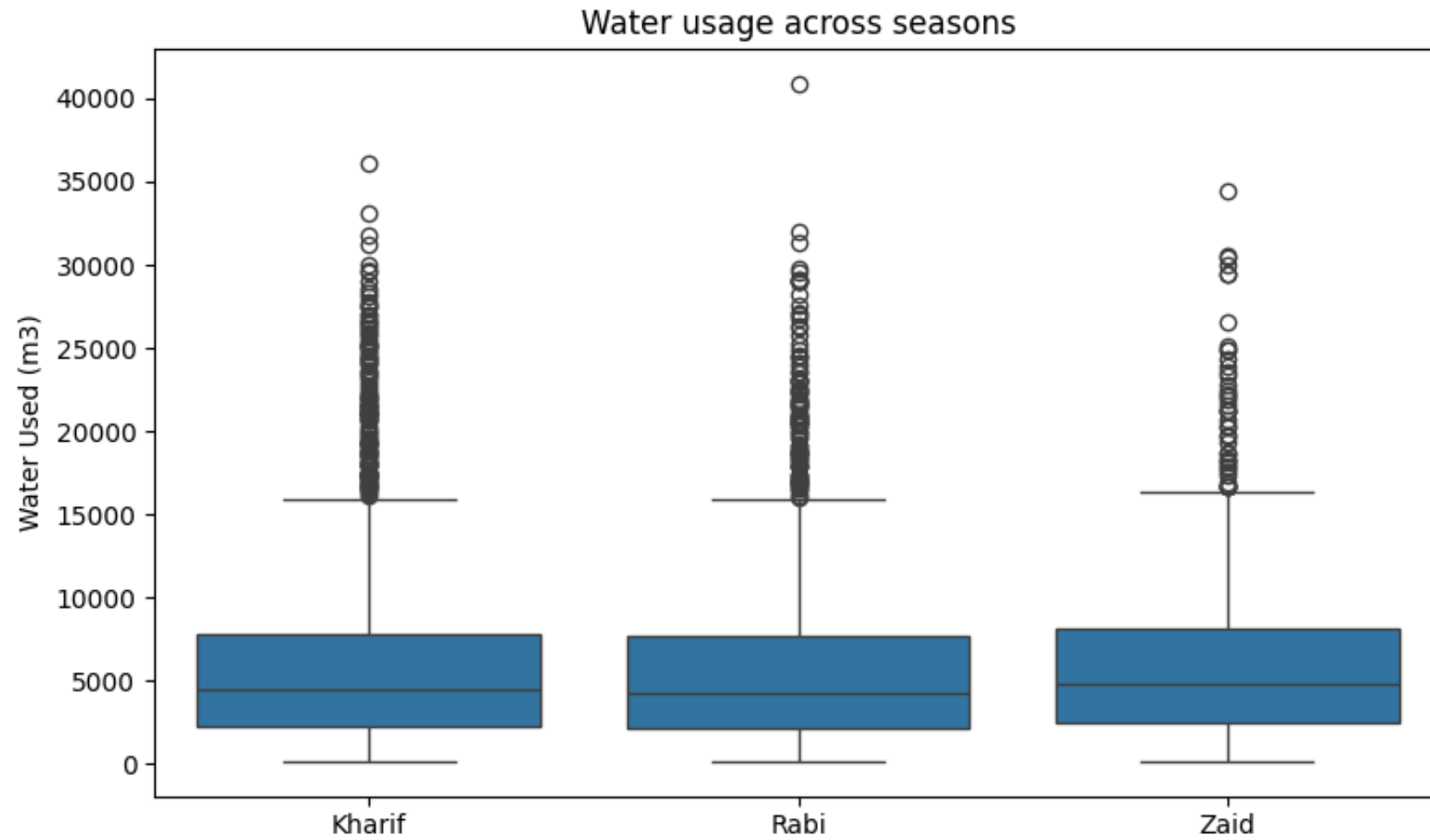
```
plt.figure(figsize=(8, 5))
sns.barplot(data=df, x="Season", y="Profit_INR", errorbar=None)
plt.title("Profit Distribution Across Seasons(Bar plot)")
plt.xlabel("Season")
```

```
plt.ylabel("Average Profit (INR)")  
plt.tight_layout()  
plt.show()
```



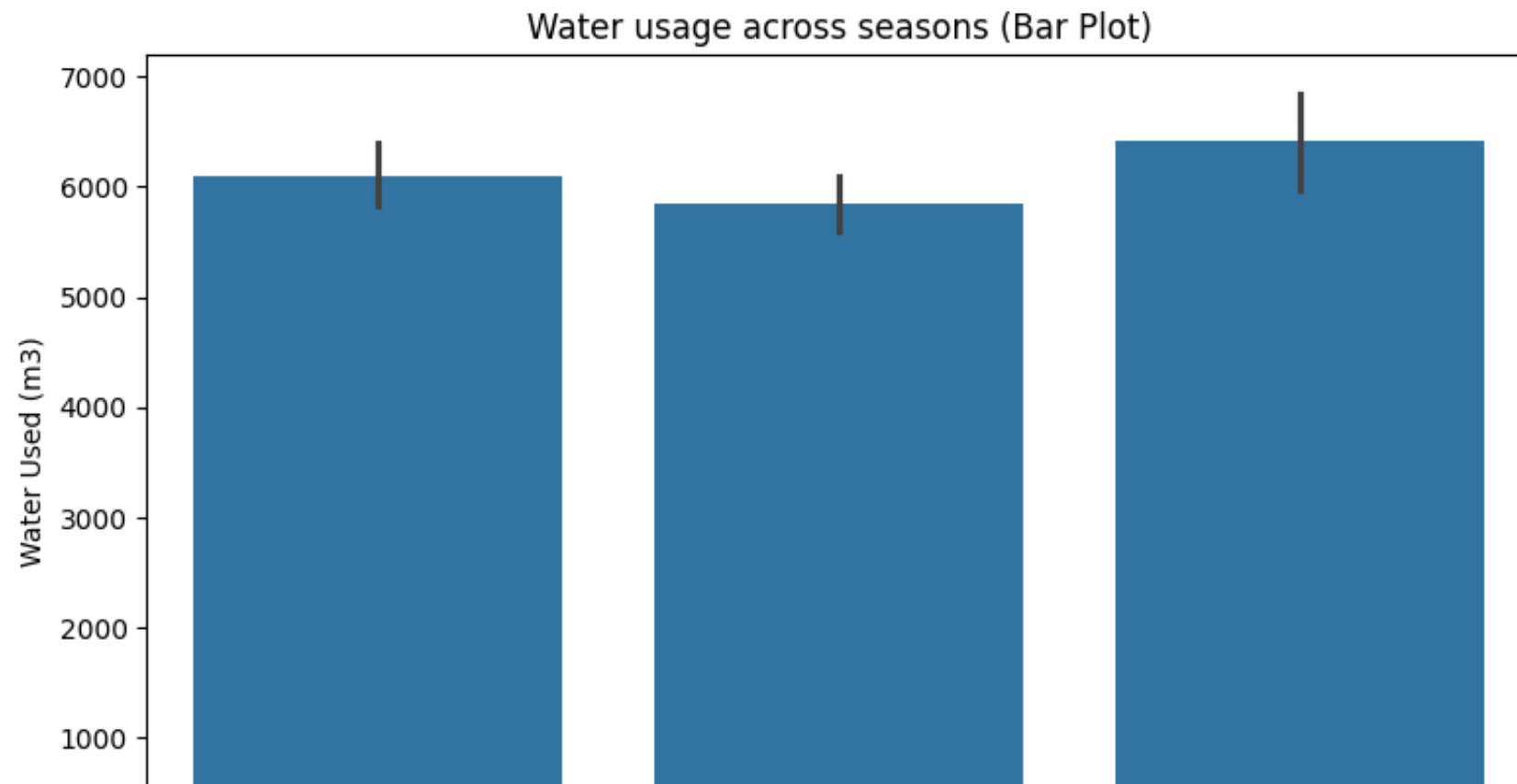
```
plt.figure(figsize=(8, 5))
```

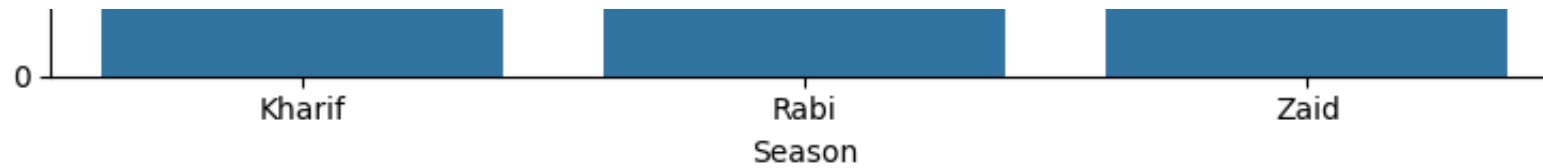
```
plt.figure(figsize=(8, 5))  
sns.boxplot(data=df, x="Season", y="Water_Used_m3")  
plt.title("Water usage across seasons")  
plt.xlabel("Season")  
plt.ylabel("Water Used (m3)")  
plt.tight_layout()  
plt.show()
```



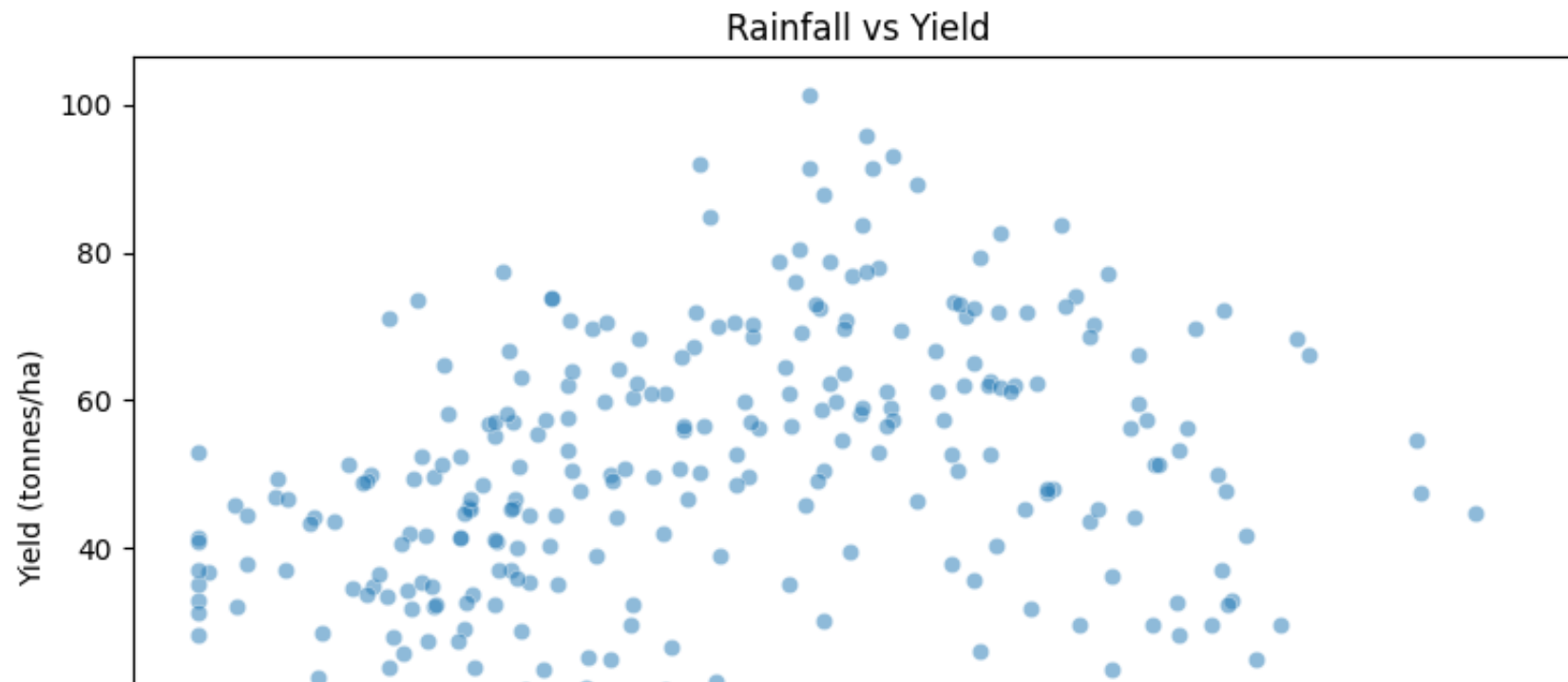
Season

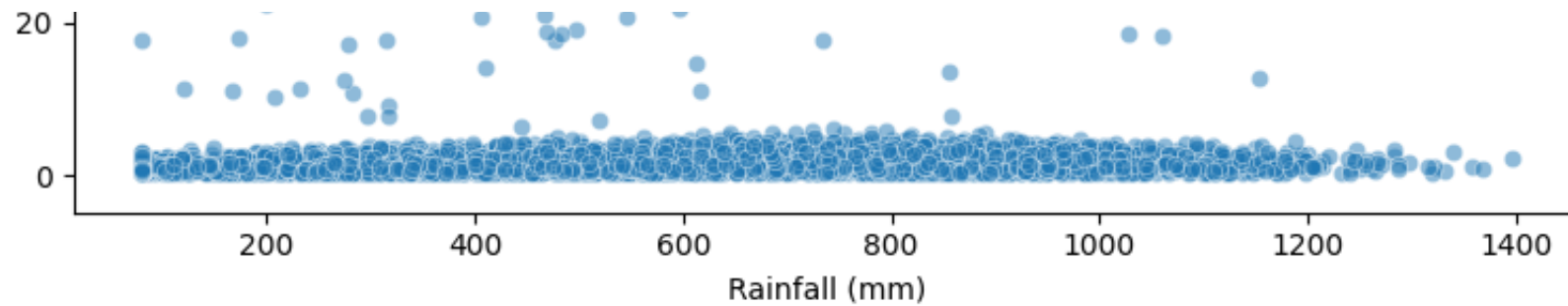
```
plt.figure(figsize=(8, 5))
sns.barplot(data=df, x="Season", y="Water_Used_m3")
plt.title("Water usage across seasons (Bar Plot)")
plt.xlabel("Season")
plt.ylabel("Water Used (m3)")
plt.tight_layout()
plt.show()
```





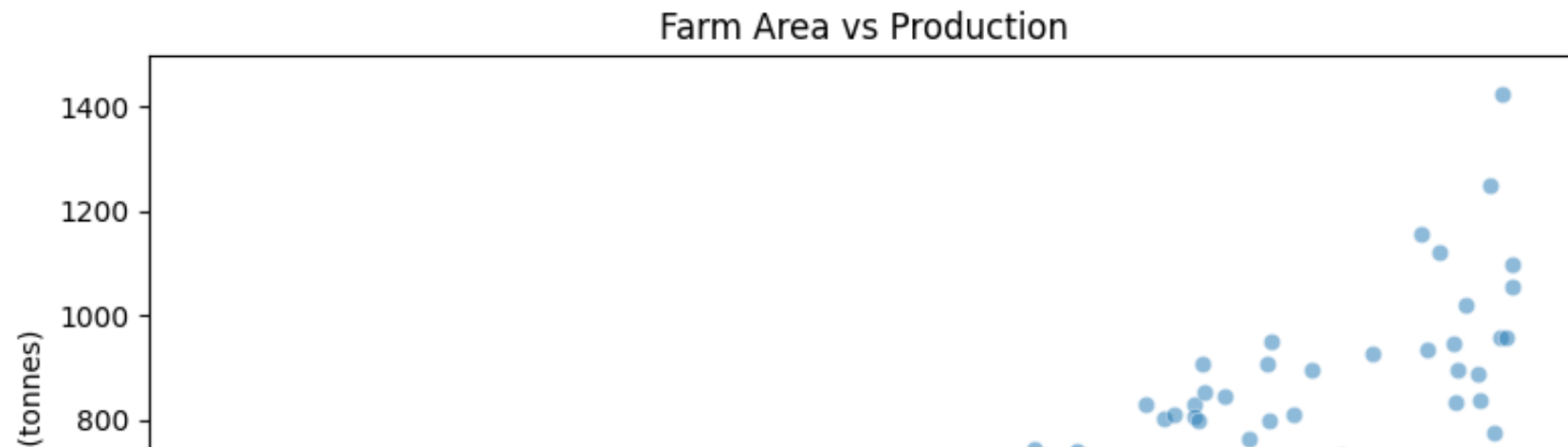
```
plt.figure(figsize=(8, 5))
sns.scatterplot(data=df, x="Rainfall_mm", y="Yield_Tonnes_Ha", alpha=0.5)
plt.title("Rainfall vs Yield")
plt.xlabel("Rainfall (mm)")
plt.ylabel("Yield (tonnes/ha)")
plt.tight_layout()
plt.show()
```

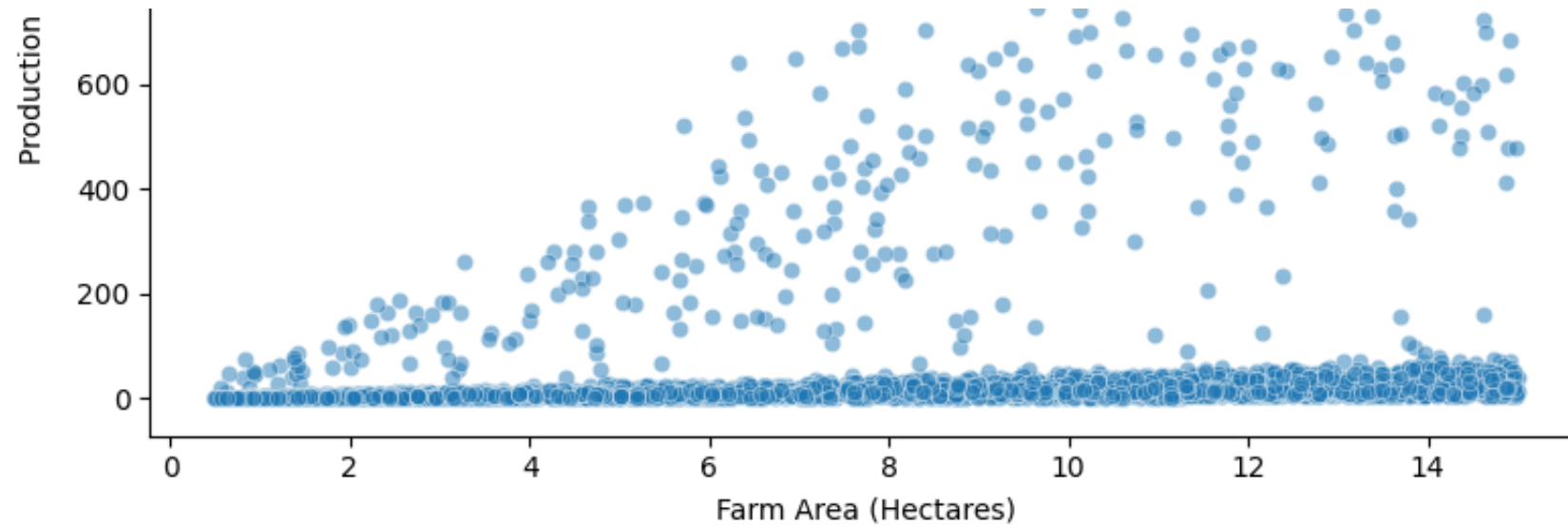




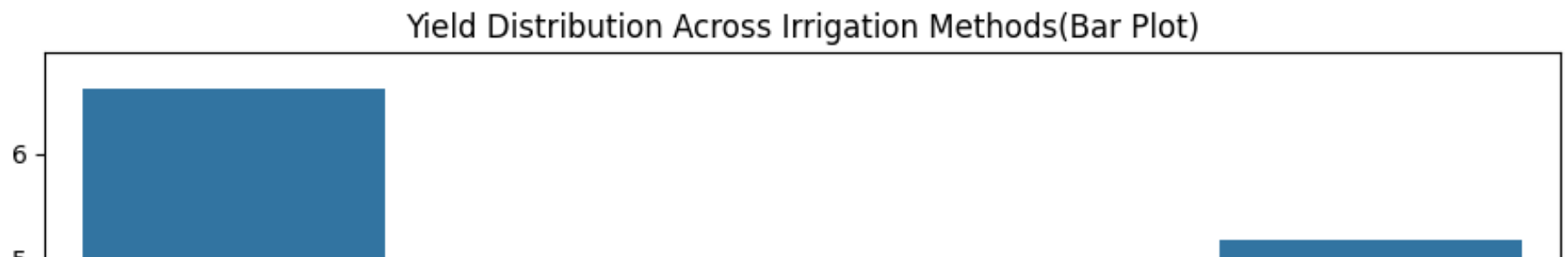
Start coding or [generate](#) with AI.

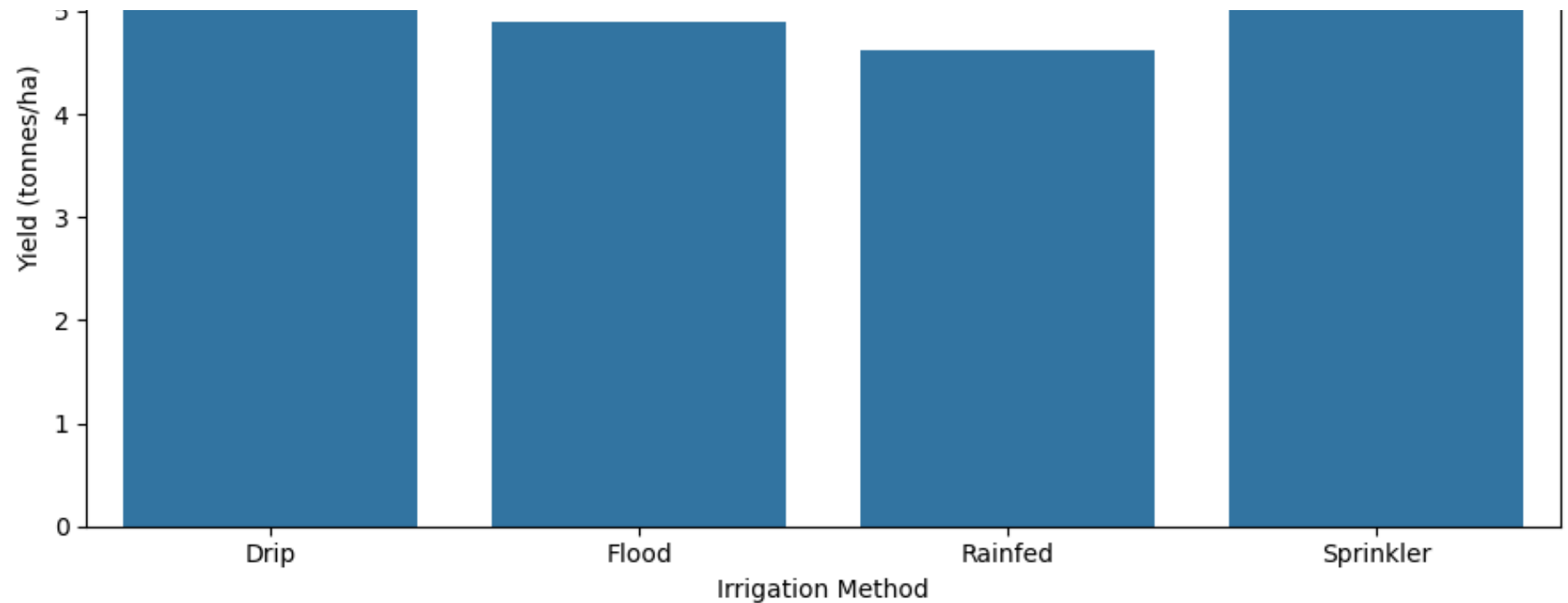
```
plt.figure(figsize=(8, 5))
sns.scatterplot(data=df, x="Farm_Area_Hectares", y="Production_Tonnes", alpha=0.5)
plt.title("Farm Area vs Production")
plt.xlabel("Farm Area (Hectares)")
plt.ylabel("Production (tonnes)")
plt.tight_layout()
plt.show()
```





```
plt.figure(figsize=(9, 5))
sns.barplot(data=df, x="Irrigation_Method", y="Yield_Tonnes_Ha", errorbar=None)
plt.title("Yield Distribution Across Irrigation Methods (Bar Plot)")
plt.xlabel("Irrigation Method")
plt.ylabel("Yield (tonnes/ha)")
plt.tight_layout()
plt.show()
```





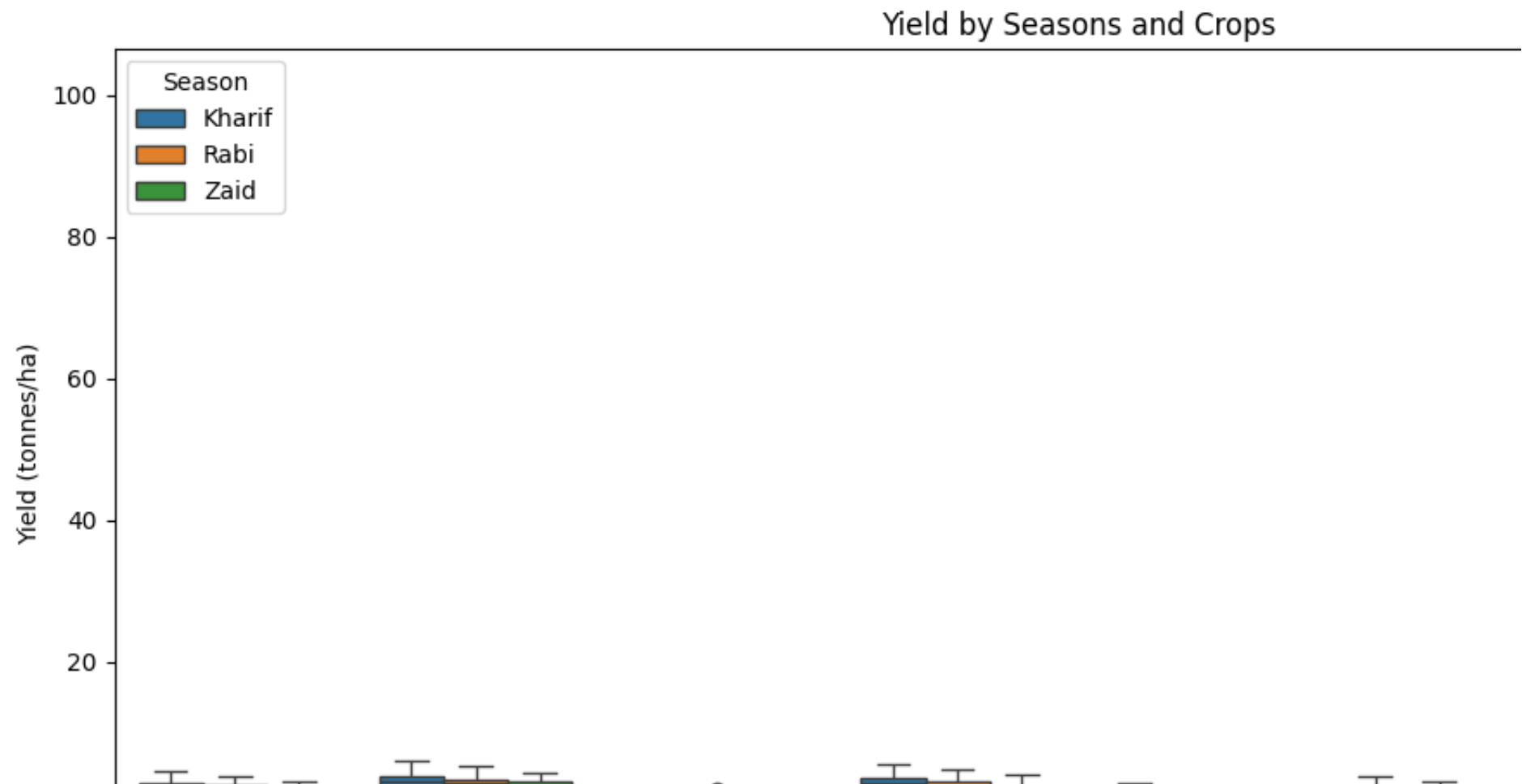
✓ 11. Multivariate Analysis

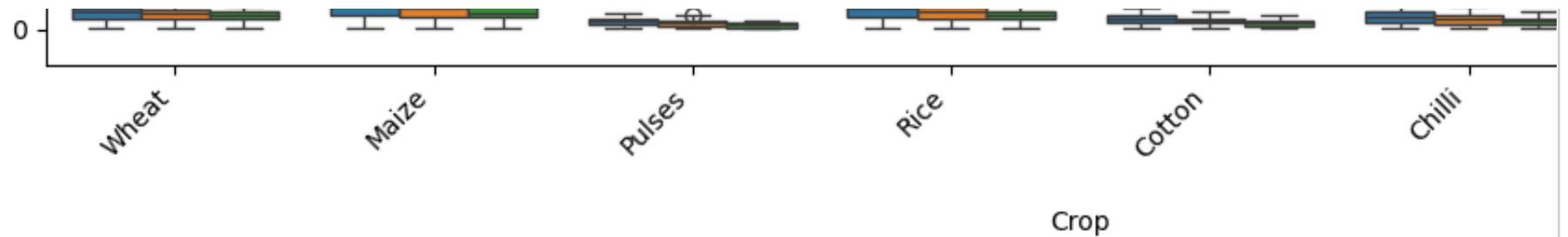
Multivariate analysis examines more than two variables together.

Use multivariate plots to investigate how season interacts with other dimensions such as crop, region, environmental conditions, resource usage or economic performance.

```
plt.figure(figsize=(12, 6))
sns.boxplot(data=df, x="Crop", y="Yield_Tonnes_Ha", hue="Season")
```

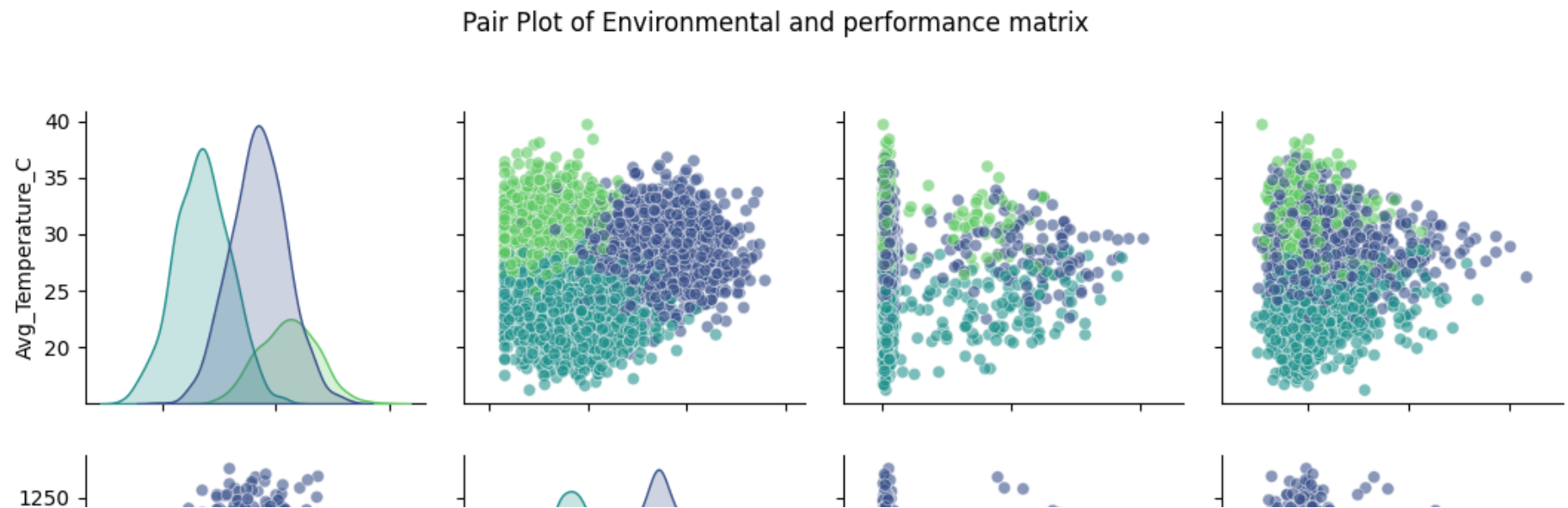
```
plt.xticks(rotation=45, ha="right")
plt.title("Yield by Seasons and Crops")
plt.xlabel("Crop")
plt.ylabel("Yield (tonnes/ha)")
plt.legend(title="Season")
plt.tight_layout()
plt.show()
```

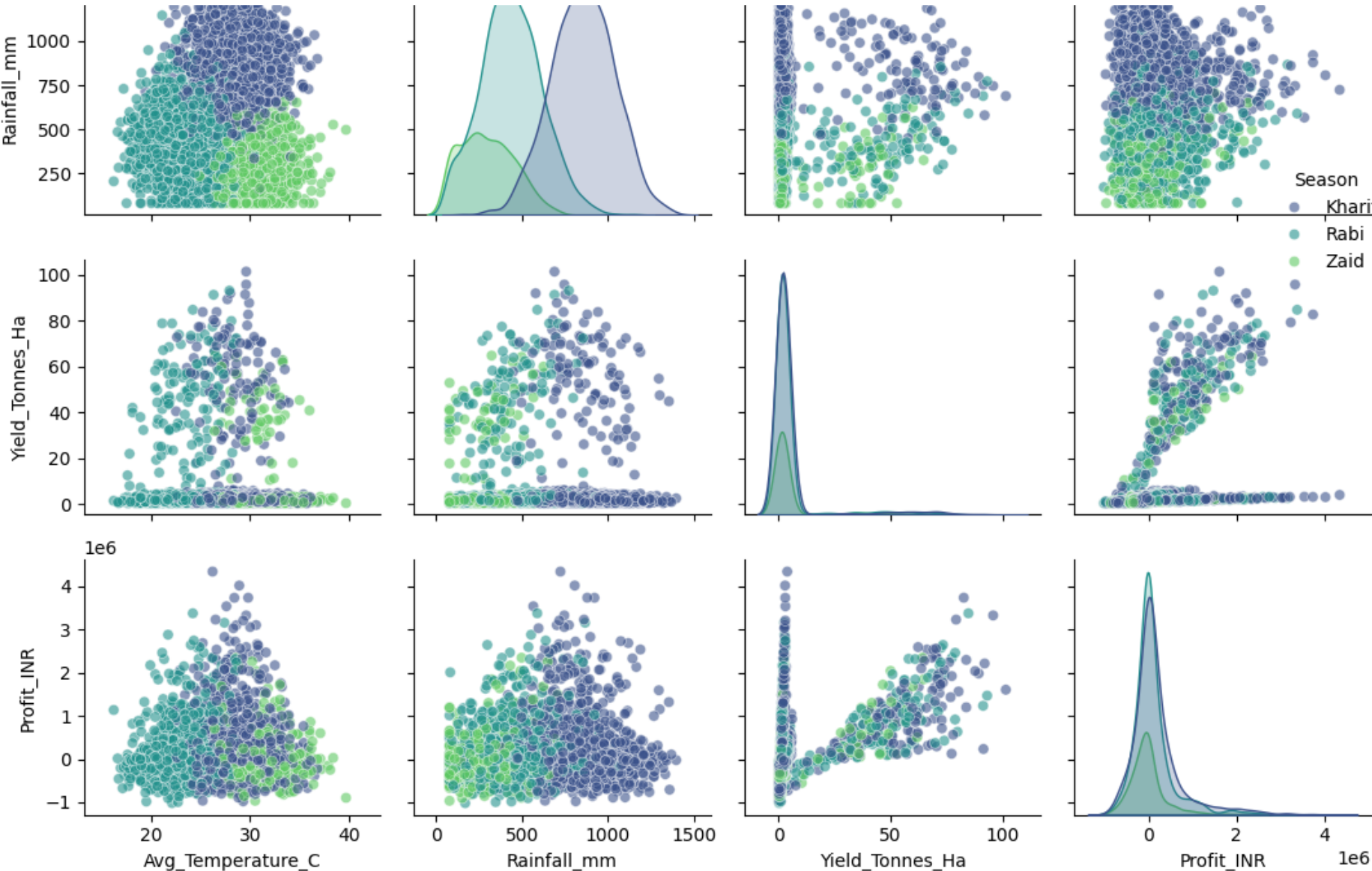




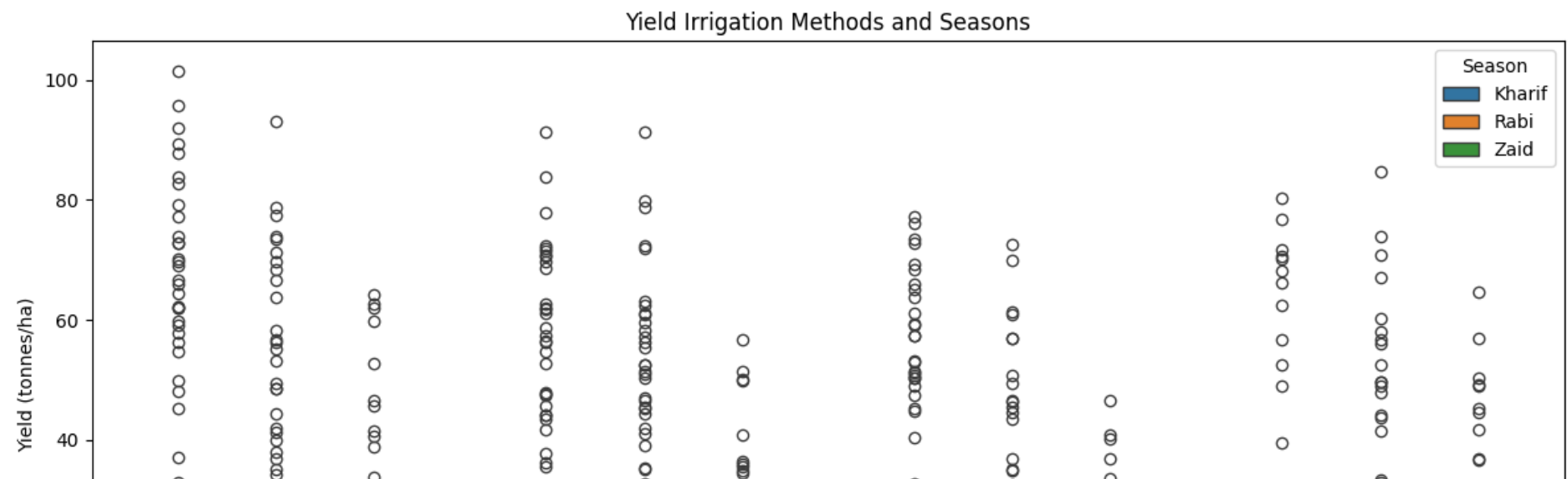
Multivariate: Pair Plot of Key Environmental and Performance Metrics by Season

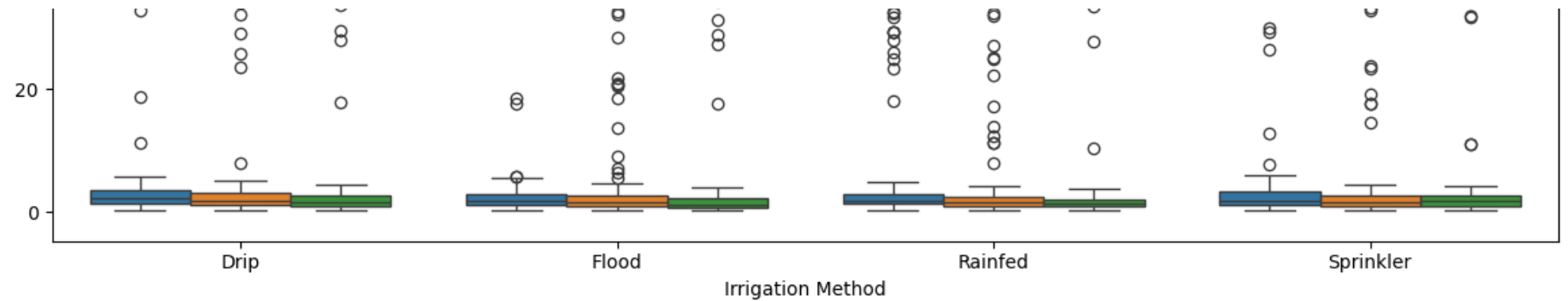
```
selected_columns = ['Avg_Temperature_C', 'Rainfall_mm', 'Yield_Tonnes_Ha', 'Profit_INR', 'Season']
sns.pairplot(df[selected_columns], hue='Season', palette='viridis', plot_kws={'alpha': 0.6})
plt.suptitle('Pair Plot of Environmental and performance matrix', y=1.02)
plt.tight_layout()
plt.show()
```



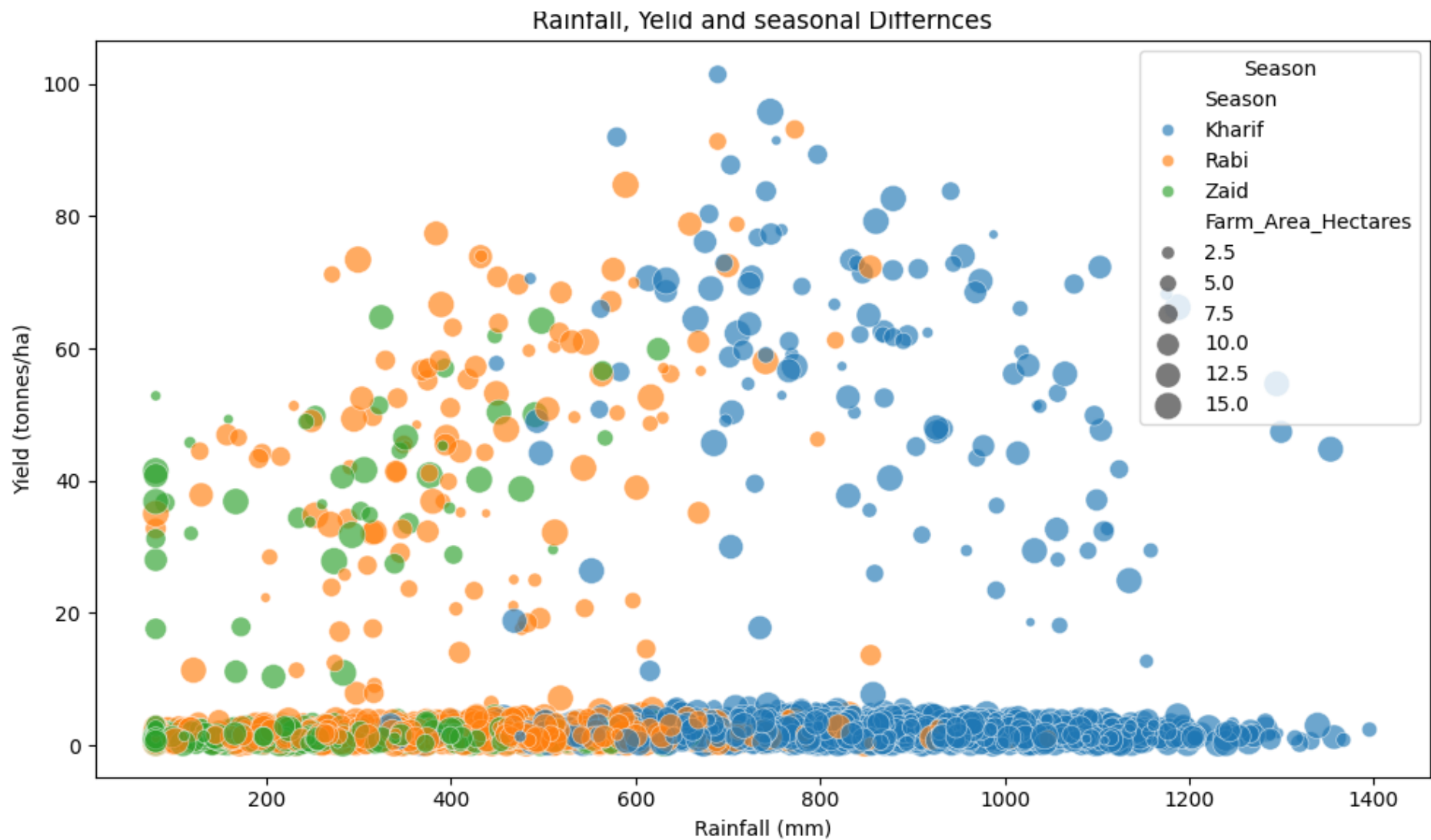


```
plt.figure(figsize=(12, 6))
sns.boxplot(data=df, x="Irrigation_Method", y="Yield_Tonnes_Ha", hue="Season")
plt.title("Yield Irrigation Methods and Seasons")
plt.xlabel("Irrigation Method")
plt.ylabel("Yield (tonnes/ha)")
plt.legend(title="Season")
plt.tight_layout()
plt.show()
```

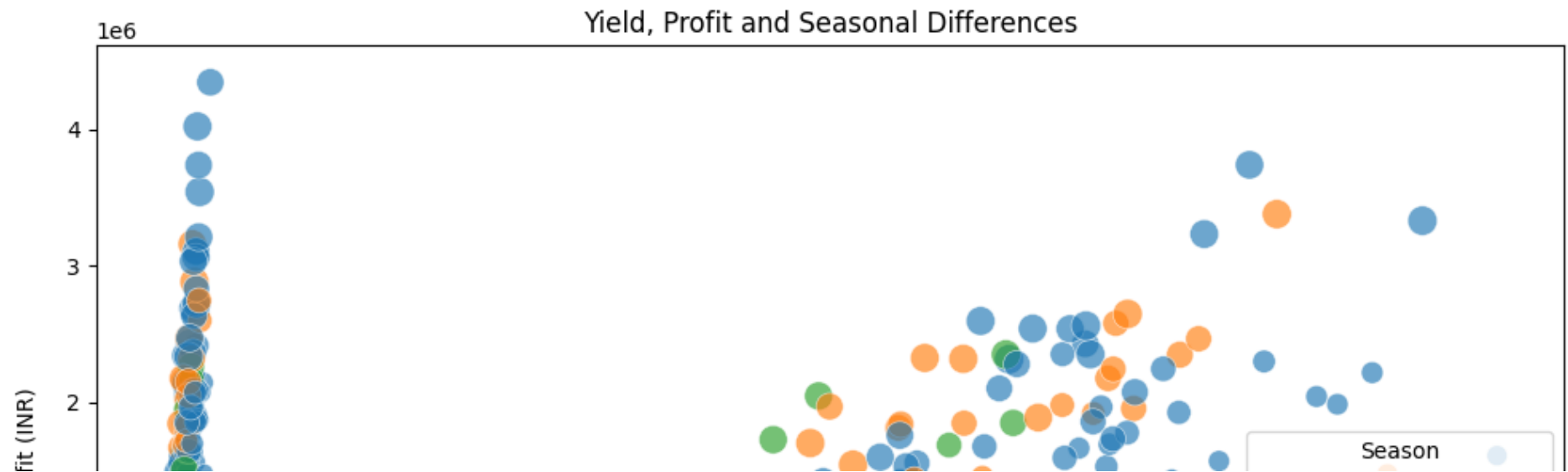


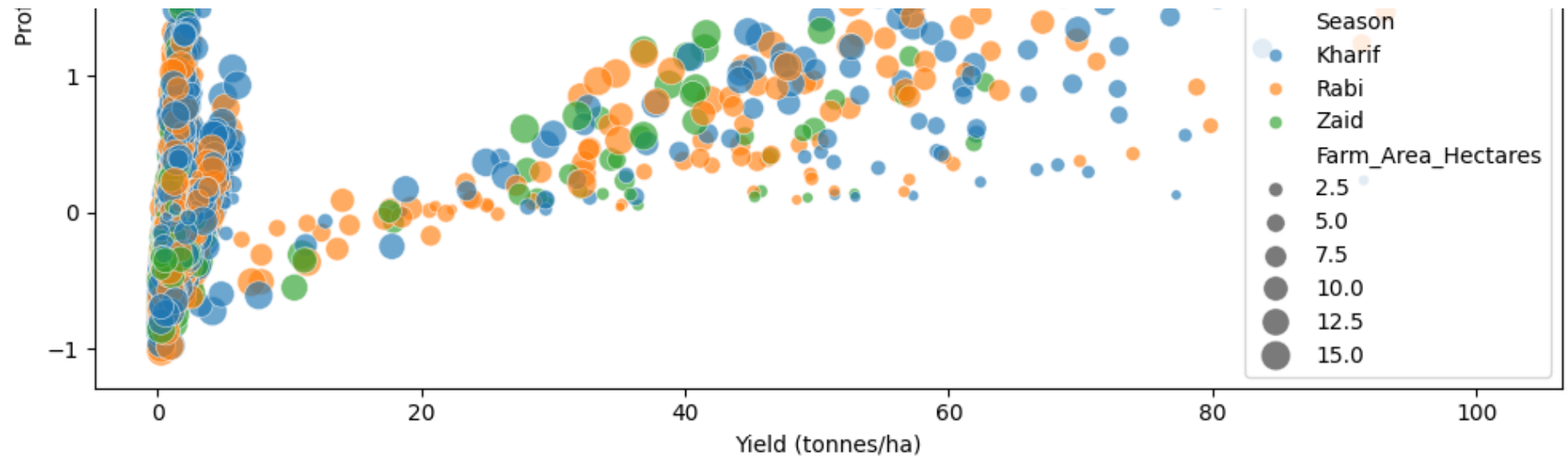


```
plt.figure(figsize=(10, 6))
sns.scatterplot(data=df, x="Rainfall_mm", y="Yield_Tonnes_Ha", hue="Season",
                size="Farm_Area_Hectares",
                sizes=(20, 180),
                alpha=0.65
                )
plt.title("Rainfall, Yield and seasonal Differences")
plt.xlabel("Rainfall (mm)")
plt.ylabel("Yield (tonnes/ha)")
plt.legend(title="Season")
plt.tight_layout()
plt.show()
```

```
plt.figure(figsize=(10, 6))
sns.scatterplot(
    data=df,
    x="Yield_Tonnes_Ha",
    y="Profit_INR",
    hue="Season",
    size="Farm_Area_Hectares",
    sizes=(20, 180),
    alpha=0.65
)
plt.title("Yield, Profit and Seasonal Differences")
plt.xlabel("Yield (tonnes/ha)")
plt.ylabel("Profit (INR)")
plt.legend(title="Season")
plt.tight_layout()
plt.show()
```





```

from pandas.core.arrays import numeric
plt.figure(figsize=(16, 12))
corr = df[numeric_columns].corr(numeric_only=True)
sns.heatmap(
    corr, annot=True,
    fmt=".2f",
    cmap="coolwarm",
    center=0,
    linewidths=0.5
)
plt.title("Correlation Matrix of Numerical Variabes")
plt.tight_layout()
plt.show()

```

#

Correlation Matrix of Numerical Variables

Farm_Area_Hectares	1.00	0.01	0.04	0.00	0.00	0.01	0.00	0.01	-0.01	0.00	0.01	-0.01	0.02
Rainfall_mm	0.01	1.00	0.20	0.52	-0.36	0.02	0.52	0.01	0.02	0.03	0.02	0.00	-0.02
Avg_Temperature_C	0.04	0.20	1.00	0.14	-0.03	0.03	0.08	0.02	0.01	0.02	0.02	0.01	-0.03
Humidity_pct	0.00	0.52	0.14	1.00	-0.30	0.03	0.45	0.00	-0.01	-0.00	0.02	-0.00	-0.05
Sunlight_Hours_Day	0.00	-0.36	-0.03	-0.30	1.00	-0.02	-0.30	0.01	0.00	-0.01	-0.01	-0.02	0.02
Soil_pH	0.01	0.02	0.03	0.03	-0.02	1.00	0.03	0.01	0.02	-0.02	-0.01	-0.01	-0.00
Soil_Moisture_pct	0.00	0.52	0.08	0.45	-0.30	0.03	1.00	0.01	0.01	0.01	0.01	-0.02	-0.02
Nitrogen_kg_ha	0.01	0.01	0.02	0.00	0.01	0.01	0.01	1.00	0.01	0.01	-0.03	0.01	0.01
Phosphorus_kg_ha	-0.01	0.02	0.01	-0.01	0.00	0.02	0.01	0.01	1.00	0.00	0.01	-0.03	-0.01
Potassium_kg_ha	0.00	0.03	0.02	-0.00	-0.01	-0.02	0.01	0.01	0.00	1.00	0.02	-0.00	0.01
Fertilizer_kg_ha	0.01	0.02	0.02	0.02	-0.01	-0.01	0.01	-0.03	0.01	0.02	1.00	-0.00	0.02
Pesticide_Litre_ha	-0.01	0.00	0.01	-0.00	-0.02	-0.01	-0.02	0.01	-0.03	-0.00	-0.00	1.00	-0.00
Seed_Quality_Score	0.02	-0.02	-0.03	-0.05	0.02	-0.00	-0.02	0.01	-0.01	0.01	0.02	-0.00	1.00

Yield_Tonnes_Ha	0.03	0.03	0.01	0.01	-0.02	-0.02	0.01	0.05	0.05	0.00	0.00	-0.01	-0.01
Production_Tonnes	0.20	0.02	0.01	0.00	-0.02	-0.01	0.00	0.05	0.03	0.00	0.00	-0.01	-0.00
Market_Price_INR_Tonne	-0.02	0.00	0.00	0.00	-0.00	0.02	0.02	-0.02	-0.01	0.01	-0.00	0.01	0.01
Total_Cost_INR	0.96	0.01	0.04	0.01	-0.00	0.01	0.00	0.01	-0.02	0.00	0.15	0.05	0.02
Revenue_INR	0.54	0.10	0.03	0.06	-0.05	-0.01	0.08	0.08	0.03	0.03	0.01	-0.00	0.02
Profit_INR	0.12	0.11	0.01	0.06	-0.05	-0.01	0.09	0.09	0.05	0.03	-0.07	-0.03	0.01
Water_Used_m3	0.58	0.00	0.02	-0.00	-0.01	-0.02	0.00	0.02	-0.00	0.00	0.00	0.00	0.02
Water_Efficiency_t_per_1000m3	0.03	0.06	0.01	0.03	-0.02	-0.01	0.03	0.05	0.06	0.01	0.01	-0.01	-0.01
Disease_Pest_Risk_pct	0.03	0.63	0.25	0.55	-0.25	0.03	0.38	-0.01	0.03	0.01	0.02	-0.47	-0.02
	Farm_Area_Hectares	Rainfall_mm	Avg_Temperature_C	Humidity_pct	Sunlight_Hours_Day	Soil_pH	Soil_Moisture_pct	Nitrogen_kg_ha	Phosphorus_kg_ha	Potassium_kg_ha	Fertilizer_kg_ha	Pesticide_Litre_ha	Seed_Quality_Score

✓ 12. Seasonal Comparison

Students should now move from individual plots to structured comparisons.

Student task

Create suitable summary tables and visualizations to compare seasons across meaningful measures.

Do not assume that every variable needs to be compared. Select variables that help answer the project problem.

```
import pandas as pd
import os

file_path = "/content/seasonal_agriculture_performance_dataset.csv"

try:
    df = pd.read_csv(file_path)

    season_metrics = df.groupby("Season").agg(
        Records=("Farm_ID", "count"),
        Average_Yield=("Yield_Tonnes_Ha", "mean"),
        Total_Production=("Production_Tonnes", "sum"),
        Average_Profit=("Profit_INR", "mean"),
        Total_Water_Used=("Water_Used_m3", "sum"),
        Average_Water_Used=("Water_Used_m3", "mean")
    ).round(2)
```

```

print(season_metrics)

except FileNotFoundError:
    print(f"Error: The file '{file_path}' was not found.")
    print("Please ensure you have uploaded 'seasonal_agriculture_performance_dataset.csv' to the Colab e
    print("You can upload it by clicking the folder icon on the left panel, then the 'Upload' icon, and
except Exception as e:
    print(f"An unexpected error occurred: {e}")

```

	Records	Average_Yield	Total_Production	Average_Profit	\
Season					
Kharif	1779	5.64	82387.61	178914.65	
Rabi	1627	5.08	67498.88	87689.47	
Zaid	594	4.67	23098.35	-24804.82	

	Total_Water_Used	Average_Water_Used
Season		
Kharif	10855816	6102.20
Rabi	9513048	5846.99
Zaid	3813417	6419.89

```

crop_season_summary = (
    df.groupby(["Crop", "Season"])
    .agg(
        Records=("Farm_ID", "count"),
        Average_Yield=("Profit_INR", "mean")
    )
    .round(2)
    .sort_values(["Season", "Average_Yield"], ascending=[True, False])
)

crop_season_summary

```

		Records	Average_Yield
Crop	Season		
Sugarcane	Kharif	125	1000790.81
Chilli	Kharif	185	954380.42
Cotton	Kharif	215	216999.55
Groundnut	Kharif	193	108265.44
Pulses	Kharif	221	43338.90
Maize	Kharif	234	-39332.67
Rice	Kharif	314	-64903.40
Wheat	Kharif	292	-105871.87
Sugarcane	Rabi	129	731612.86
Chilli	Rabi	163	638572.50
Cotton	Rabi	201	107343.59
Groundnut	Rabi	177	10416.54
Pulses	Rabi	213	-14309.14
Maize	Rabi	233	-89133.40
Rice	Rabi	274	-98427.87
Wheat	Rabi	237	-119954.57
Total		51	500005.00

Sugarcane	Zaid	51	583635.80
Chilli	Zaid	64	448659.09
Cotton	Zaid	92	-53925.34
Groundnut	Zaid	54	-68872.50
Pulses	Zaid	62	-139227.81
Wheat	Zaid	85	-193208.95
Maize	Zaid	84	-194049.17
Rice	Zaid	102	-227239.34

```

import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

print("\n--- Analysis 1: Regional Differences in Crop Yield by Season ---")
state_crop_season_yield = df.groupby(['State', 'Crop', 'Season'])['Yield_Tonnes_Ha'].mean().unstack()
print("Average Yield (Tonnes/Ha) by State, Crop, and Season (Top 10 combinations):\n", state_crop_season_

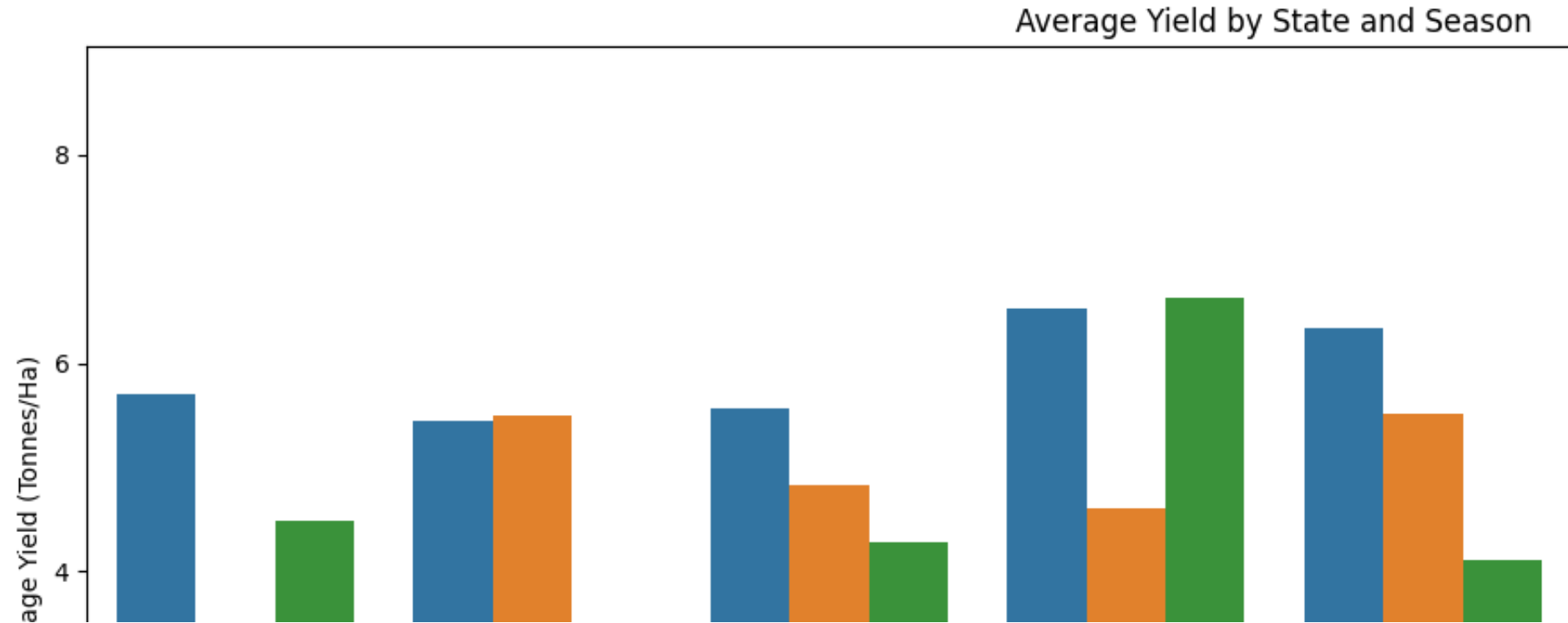
plt.figure(figsize=(14, 7))
sns.barplot(data=df, x='State', y='Yield_Tonnes_Ha', hue='Season', estimator=np.mean, errorbar=None)
plt.title('Average Yield by State and Season')
plt.xlabel('State')
plt.ylabel('Average Yield (Tonnes/Ha)')
plt.xticks(rotation=45, ha='right')
plt.legend(title='Season')
plt.tight_layout()
plt.show()

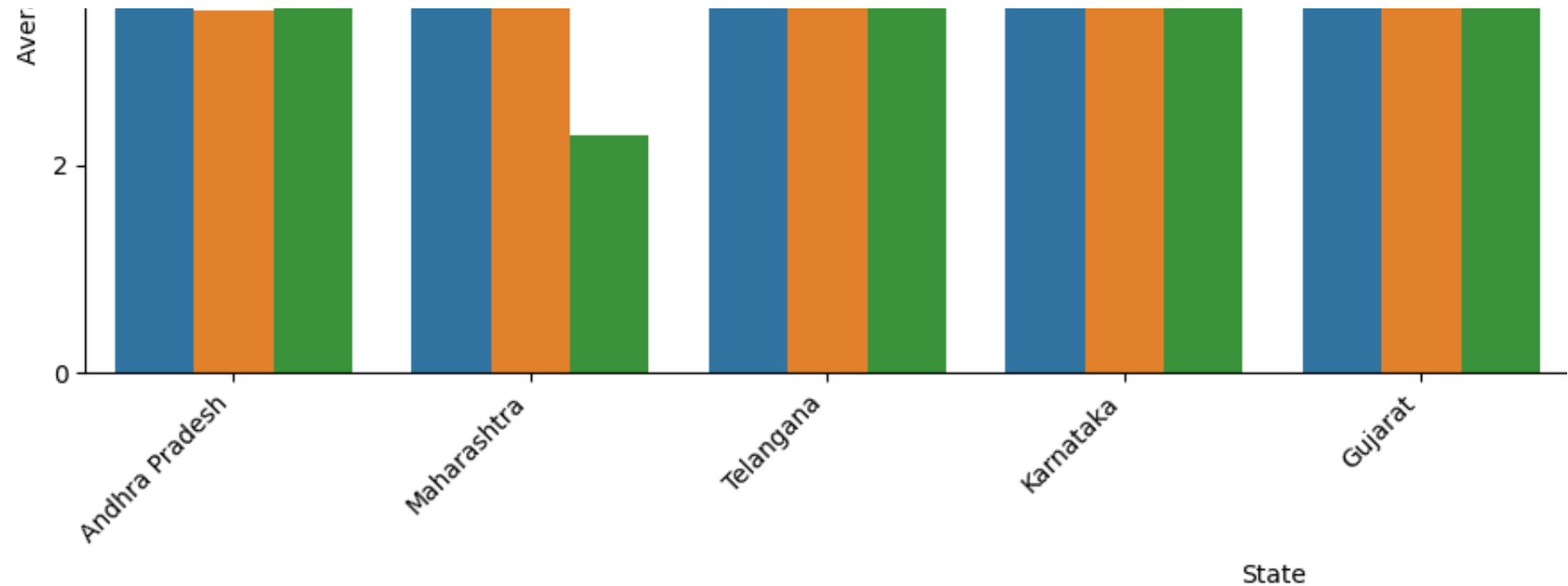
```

--- Analysis 1: Regional Differences in Crop Yield by Season ---

Average Yield (Tonnes/Ha) by State, Crop, and Season (Top 10 combinations):

Season		Kharif	Rabi	Zaid
State	Crop			
Andhra Pradesh	Chilli	1.735000	1.255000	0.841429
	Cotton	1.434348	1.138437	0.971176
	Groundnut	1.575517	1.161563	0.874000
	Maize	3.034138	2.643333	2.568333
	Pulses	1.153478	0.832333	0.485000
	Rice	2.790962	2.399565	2.041053
	Sugarcane	55.586250	51.310000	36.901429
Gujarat	Wheat	2.125116	1.590323	1.704545
	Chilli	1.817500	1.495882	1.238000
	Cotton	1.274516	1.209524	0.877143

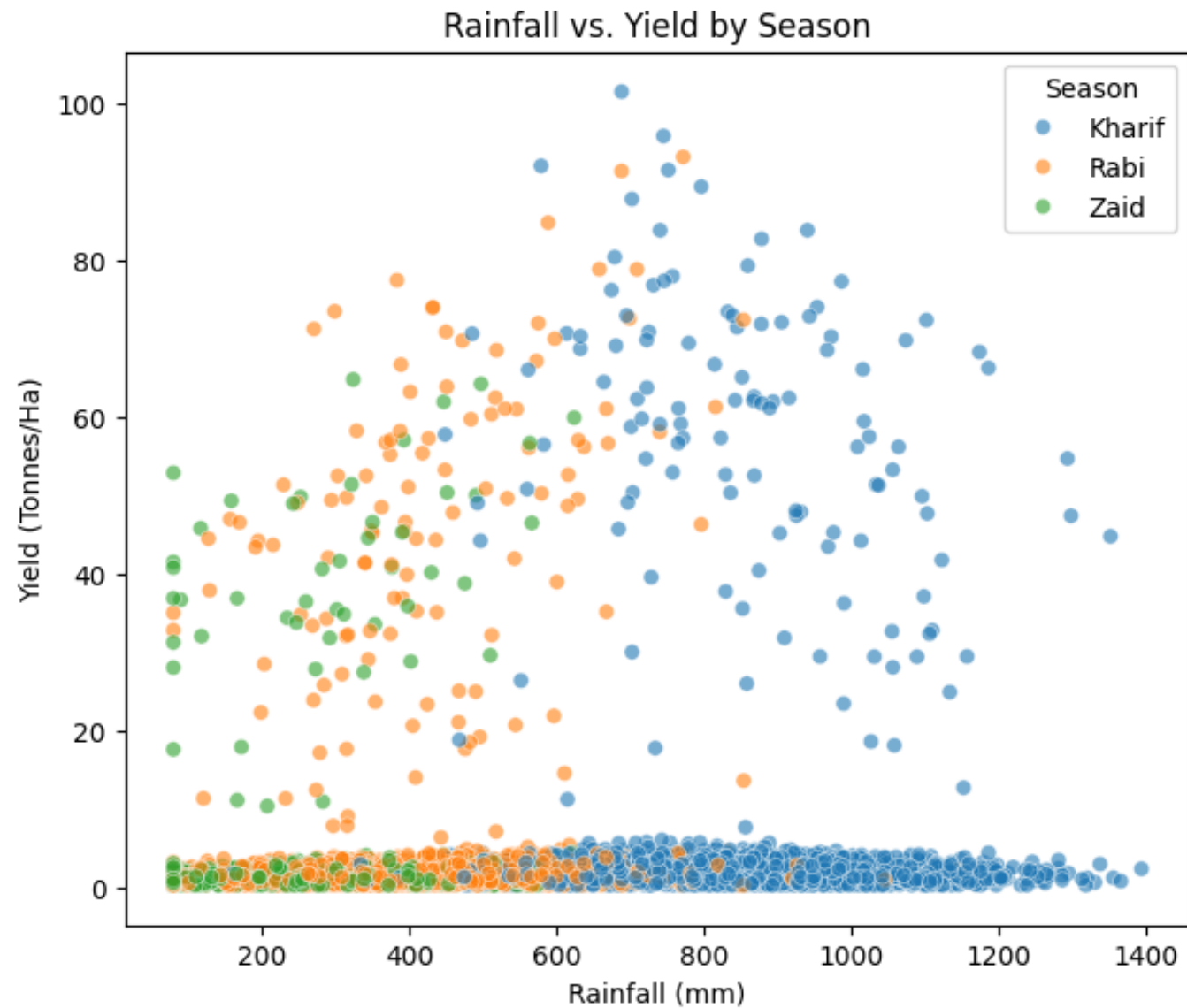




```
print("\n--- Analysis 2: Relationship between Environmental Factors and Yield across Seasons ---")
plt.figure(figsize=(16, 6))
```

```
plt.subplot(1, 2, 1)
sns.scatterplot(data=df, x='Rainfall_mm', y='Yield_Tonnes_Ha', hue='Season', alpha=0.6)
plt.title('Rainfall vs. Yield by Season')
plt.xlabel('Rainfall (mm)')
plt.ylabel('Yield (Tonnes/Ha)')
```

```
--- Analysis 2: Relationship between Environmental Factors and Yield across Seasons ---
Text(0, 0.5, 'Yield (Tonnes/Ha)')
```



```
print("\n--- Analysis 3: Profitability by Irrigation Method and Season ---")
```

```

irrigation_profit_season = df.groupby(['Irrigation_Method', 'Season'])['Profit_INR'].mean().unstack()
print("Average Profit (INR) by Irrigation Method and Season:\n", irrigation_profit_season)

plt.figure(figsize=(12, 7))
sns.barplot(data=df, x='Irrigation_Method', y='Profit_INR', hue='Season', estimator=np.mean, errorbar=None)
plt.title('Average Profit by Irrigation Method and Season')
plt.xlabel('Irrigation Method')
plt.ylabel('Average Profit (INR)')
plt.legend(title='Season')
plt.tight_layout()
plt.show()

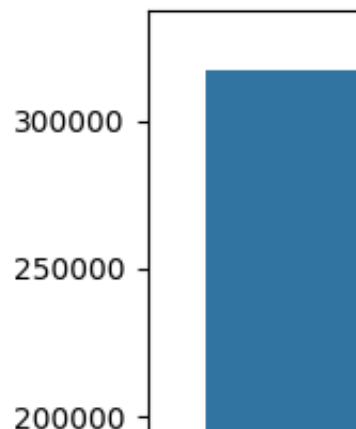
```

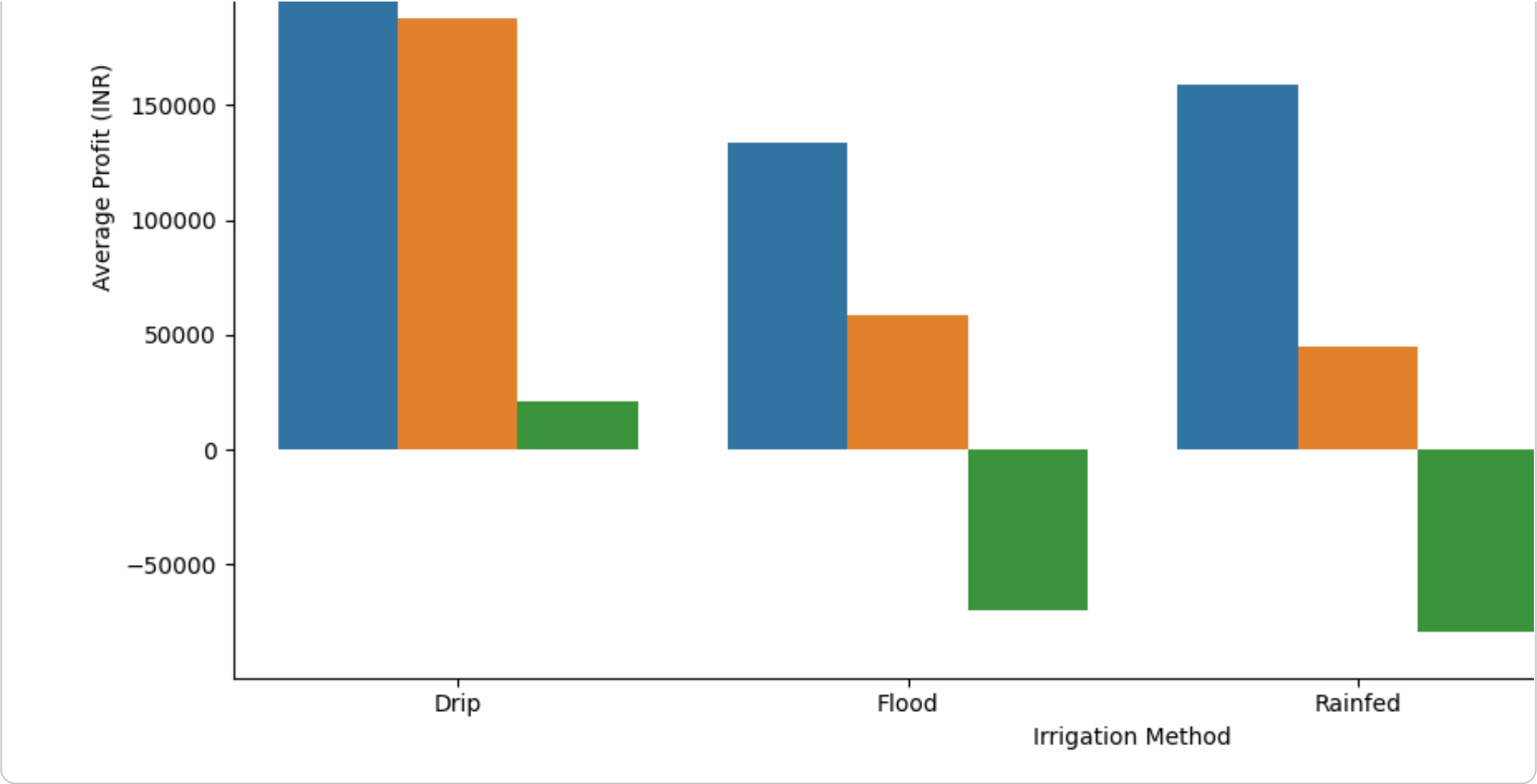
--- Analysis 3: Profitability by Irrigation Method and Season ---

Average Profit (INR) by Irrigation Method and Season:

Season	Kharif	Rabi	Zaid
Irrigation_Method			
Drip	317222.071605	187843.221622	21291.835714
Flood	133373.832203	58824.798479	-69786.778351
Rainfed	158974.125270	45232.479452	-79467.257143
Sprinkler	116880.492212	76502.068259	57909.400000

Average Profit by Irrigation Method and Season





KEY INSIGHTS

1. Insight: Seasonal Profitability Variations

1. **Observation:** The Zaid season consistently exhibits an average financial loss across all agricultural activities recorded, while the Kharif season is the most profitable.
2. **Evidence:** The `season_metrics` table (from `_na__Ekemz3H`) shows an average profit of -INR 24,804.82 for Zaid season, compared to INR 178,914.65 for Kharif and INR 87,689.47 for Rabi. The bar plot 'Profit Distribution Across Seasons(Bar plot)' visually supports this by showing negative average profit for Zaid.
3. **Interpretation:** The Zaid season poses significant economic challenges for farmers, suggesting higher operational costs, lower market prices, or reduced yields that lead to overall unprofitability on average. Conversely, Kharif is the most economically favorable season.
4. **Limitation:** This is an aggregated average across all crops and regions. Certain niche crops or specific regional practices might still be profitable during the Zaid season, which this high-level analysis doesn't disaggregate.

2. Insight: Crop-Specific Seasonal Profitability

1. **Observation:** Sugarcane and Chilli are consistently the most profitable crops across all seasons, while staple crops like Wheat, Maize, Pulses, and Rice frequently lead to financial losses, especially during the Zaid season.
2. **Evidence:** The `crop_season_summary` table (from `2MNe0u-XoB-a`) explicitly lists Sugarcane and Chilli with the highest average profits in Kharif, Rabi, and Zaid seasons. For example, Sugarcane in Kharif shows an average profit of 1,000,790.81 INR. Conversely, Wheat, Maize, Rice, and Pulses show negative average profits in most seasons, with significant losses in Zaid (e.g., Wheat Zaid: -193,208.95 INR).
3. **Interpretation:** Cultivating high-value crops like Sugarcane and Chilli appears to be a more financially viable strategy regardless of the season. Farmers should exercise caution or implement specific strategies when growing staple grains during Zaid, as they tend to be unprofitable.
4. **Limitation:** This insight is based on average profitability. Local market demand, specific input costs, and unique farm-level efficiencies for these crops could vary, impacting actual profits for individual farmers.

3. Insight: Kharif Season Dominance in Agricultural Activity and Production

1. **Observation:** The Kharif season accounts for the highest number of recorded agricultural activities and significantly higher total production compared to Rabi and Zaid seasons.
2. **Evidence:** The `season_metrics` table (from `_na__Ekemz3H`) shows Kharif with 1,779 records and a total production of 82,387.61 tonnes. This is considerably higher than Rabi (1,627 records, 67,498.88 tonnes) and Zaid (594 records, 23,098.35 tonnes). The count plot 'Distribution of Records by Seasons' also visually confirms Kharif having the largest share of records.
3. **Interpretation:** Kharif is the most active and productive agricultural season, likely coinciding with the monsoon period, which supports extensive cultivation. This makes it a critical period for overall agricultural output.
4. **Limitation:** Higher production volume does not automatically equate to higher efficiency or profitability per unit. It could reflect a larger area under cultivation or more widespread participation during this season.

4. Insight: Zaid Season's Intensive Water Usage

1. **Observation:** Despite lower overall agricultural activity, the Zaid season has the highest average water usage per farm compared to Kharif and Rabi seasons.
2. **Evidence:** The `season_metrics` table (from `_na__Ekemz3H`) reveals that Zaid season has an average water usage of 6,419.89 m³ per farm, which is higher than Kharif (6,102.20 m³) and Rabi (5,846.99 m³). The bar plot 'Water usage across seasons' further illustrates this trend.
3. **Interpretation:** The Zaid season's characteristic hot and dry conditions likely necessitate more intensive irrigation to sustain crops, leading to higher average water consumption per farm. This increased resource demand could be a contributing factor to its observed lower profitability.
4. **Limitation:** The average water usage doesn't account for the types of crops grown in Zaid, which might inherently be more water-intensive. It also doesn't differentiate between efficient and inefficient water use practices.

5. Insight: Drip Irrigation's Economic Advantage

1. **Observation:** Drip irrigation consistently yields the highest average profits across all seasons when compared to Flood, Rainfed, and Sprinkler irrigation methods.
2. **Evidence:** The `irrigation_profit_season` table and the 'Average Profit by Irrigation Method and Season' bar plot (from `UAmxhby4-lWb`) clearly show Drip irrigation outperforming other methods in average profit (e.g., Kharif: 317,222.07 INR, Rabi: 187,843.22 INR, Zaid: 21,291.84 INR). Flood and Rainfed methods even show average losses during the Zaid season.
3. **Interpretation:** Drip irrigation appears to be the most economically efficient irrigation method. This could be due to optimized water use leading to better yields, reduced water-related costs, or both, making it a preferable choice for maximizing financial returns.
4. **Limitation:** Correlation does not imply causation. Farms utilizing drip irrigation might also employ other modern agricultural practices, cultivate higher-value crops, or benefit from favorable microclimates, all of which could contribute to higher profitability.

6. Insight: Positive Association between Rainfall and Yield

1. **Observation:** There is a discernible positive association between rainfall levels and crop yield, particularly prominent in the Kharif season, although excessive rainfall does not guarantee proportional yield increases.
2. **Evidence:** The scatter plot 'Rainfall vs. Yield by Season' (from `WW23gCZJ-dxB`) illustrates that higher rainfall amounts often correspond to higher crop yields, especially for Kharif season observations. However, the dispersion of points suggests that beyond an optimal range, increasing rainfall doesn't linearly translate to higher yields.
3. **Interpretation:** Adequate rainfall is a crucial environmental factor for achieving good crop yields. The variability observed suggests that while essential, there might be an optimal rainfall window, and extreme amounts could introduce other challenges affecting yield.

4. **Limitation:** This analysis indicates an association. The specific impact of rainfall on yield is highly dependent on crop type, soil drainage, temperature, and other environmental factors. It doesn't isolate rainfall as the sole causal factor for yield variations.

7. Insight: Environmental Characterization of Zaid Season

1. **Observation:** The Zaid season is predominantly characterized by lower rainfall and higher average temperatures compared to the Kharif and Rabi seasons.
2. **Evidence:** The `season_metrics` table (from `_na__Ekemz3H`) shows Zaid with the lowest average rainfall. The scatter plots in 'Relationship between Environmental Factors and Yield across Seasons' (from `WW23gCZJ-dxB`) visually group Zaid observations in lower rainfall and higher temperature regions. The 'Pair Plot of Environmental and performance matrix' (`YsWeo65DhzlF`) further reinforces this environmental distinction for Zaid.
3. **Interpretation:** The harsh environmental conditions of Zaid—less rainfall and higher temperatures—create a challenging growing environment. These conditions likely increase the reliance on irrigation and contribute to the overall lower yields and profitability observed during this season.
4. **Limitation:** These are average environmental conditions. Specific microclimatic variations within different regions and potential extreme weather events are not captured, which could significantly impact localized agricultural outcomes.

8. Insight: Strong Correlation Between Yield and Profit

1. **Observation:** There is a strong positive correlation between the Yield (Tonnes/Ha) and the Profit (INR).
2. **Evidence:** The 'Correlation Matrix of Numerical Variables' (from `KnGT-eSWltGS`) shows a high positive correlation coefficient between `Yield_Tonnes_Ha` and `Profit_INR`. Visually, the scatter plot 'Yield, Profit and Seasonal Differences' (from `RW_hCbSq1KQ2`) demonstrates a clear upward trend: as yield increases, profit generally increases across all seasons.

3. **Interpretation:** Maximizing yield per hectare is a key determinant of farm profitability. Efforts and investments aimed at improving crop yield are likely to translate directly into enhanced economic returns for farmers.
4. **Limitation:** While highly correlated, profit is also influenced by other significant factors such as market prices, total operational costs, and the efficiency of resource utilization. High yields may not guarantee high profits if market prices are low or production costs are excessively high.

✓ Recommendations

1. Recommendation: Strategic Crop Planning for Zaid Season

- **Recommendation:** Farmers should critically evaluate crop choices for the Zaid season, prioritizing high-value crops like Sugarcane and Chilli, and consider alternative strategies for staple grains or avoid them if local conditions do not support profitability.
- **Justification (based on Insight 1 & 2):** The Zaid season consistently shows average financial losses, and staple crops like Wheat, Maize, Rice, and Pulses are often unprofitable during this period. Conversely, Sugarcane and Chilli demonstrate strong profitability across all seasons, including Zaid.
- **Limitation:** This recommendation is based on average profitability. Local demand, specific input costs, and unique farm-level efficiencies for these crops may vary. Detailed local market analysis and cost-benefit studies are still necessary for individual farmers.

2. Recommendation: Promote Drip Irrigation Adoption

- **Recommendation:** Agricultural extension services should actively promote and subsidize the adoption of drip irrigation

systems, especially in regions with lower rainfall or during drier seasons.

- **Justification (based on Insight 4 & 5):** Drip irrigation consistently leads to higher average profits across all seasons, suggesting better resource efficiency and economic returns. The Zaid season, despite having lower activity, shows the highest average water usage, indicating a need for efficient irrigation.
- **Limitation:** While drip irrigation shows a strong association with higher profits, causality is not definitively proven by this analysis alone. Farms adopting drip irrigation might also have other advanced practices or favorable conditions that contribute to profitability. Initial investment costs for drip systems can be a barrier for some farmers.

3. Recommendation: Targeted Support for Kharif Season Optimization

- **Recommendation:** Government and agricultural bodies should focus on optimizing support (e.g., improved seeds, market access, weather advisories) for the Kharif season, given its significant contribution to overall agricultural output.
- **Justification (based on Insight 3):** The Kharif season accounts for the highest number of agricultural records and the largest total production, making it the most impactful season for national agricultural output.
- **Limitation:** While Kharif is dominant in production, focusing solely on this season might neglect potential for growth or improved resilience in other seasons. 'Support' is a broad term and specific interventions need to be tailored to actual needs.

4. Recommendation: Develop Drought-Resistant Strategies for Zaid Season

- **Recommendation:** Research and development efforts should focus on cultivating drought-resistant crop varieties suitable for the Zaid season and implementing water-saving agricultural practices tailored to hot and dry conditions.
- **Justification (based on Insight 7):** The Zaid season is characterized by lower rainfall and higher average temperatures, which likely contribute to its intensive water usage and overall unprofitability. Developing crops and practices resilient to these conditions could improve viability.
- **Limitation:** Implementing new crop varieties or practices requires significant lead time, farmer education, and

investment. Success depends on local adaptability and farmer acceptance.

5. Recommendation: Invest in Yield-Enhancing Technologies and Practices

- **Recommendation:** Farmers should be encouraged and supported to adopt technologies and practices proven to enhance crop yield, as this has a direct positive impact on profitability.
- **Justification (based on Insight 8):** There is a strong positive correlation between `Yield_Tonnes_Ha` and `Profit_INR`, indicating that higher yields generally lead to higher profits.
- **Limitation:** While yield is strongly correlated with profit, other factors such as market price fluctuations, pest/disease outbreaks, and input costs can also significantly influence final profitability. Increasing yield should not come at the expense of unsustainably high costs.

6. Recommendation: Regionalized Crop and Seasonal Advisories

- **Recommendation:** Provide region-specific advisories that detail optimal crop choices and cultivation practices for each season, considering state-level yield performance.
- **Justification (based on Insight 1):** Significant regional differences exist in crop yields across seasons. For example, specific State-Crop combinations show varying yields in Kharif, Rabi, and Zaid.
- **Limitation:** This recommendation requires granular data and localized expertise. General state-level advice may not capture microclimatic or soil variations within districts.

7. Recommendation: Integrated Pest and Disease Management for Kharif

- **Recommendation:** Implement robust integrated pest and disease management (IPM) strategies, particularly during the Kharif season, to mitigate risks and protect high production volumes.
- **Justification (based on Insight 3 & environmental variables related to humidity/rainfall):** Kharif is the most active and productive season. High humidity and rainfall often associated with Kharif create favorable conditions for pests and

diseases, which, if uncontrolled, can significantly impact the high production volumes.

- **Limitation:** The current analysis shows 'Disease_Pest_Risk_pct' as a variable, but direct evidence of its impact on yield or profit by season in the provided insights is not explicitly detailed. This recommendation infers risk based on general agricultural knowledge combined with seasonal conditions.

8. Recommendation: Water Usage Efficiency Monitoring and Education

- **Recommendation:** Develop programs to educate farmers on best practices for water usage efficiency and provide tools for monitoring water consumption across seasons.
- **Justification (based on Insight 4):** Despite lower activity, the Zaid season has the highest average water usage per farm. This indicates a potential for inefficient water use or the cultivation of water-intensive crops in challenging conditions.
- **Limitation:** The analysis indicates average water usage but doesn't distinguish between necessity and inefficiency. Actual water-saving potential varies based on crop type, soil, and existing irrigation infrastructure. Education alone may not be sufficient without accessible and affordable technology.

Conclusion

This analysis provided a deep dive into the `seasonal_agriculture_performance_dataset`, revealing critical insights into how agricultural performance varies across different seasons in India.

What the Dataset Revealed About Seasonal Agricultural Performance

The dataset clearly demonstrated that agricultural performance is highly heterogeneous across seasons. The **Kharif season** emerged as the most dominant in terms of activity and total production, generally exhibiting the highest average profitability.

The **Rabi season** showed moderate activity and profitability, while the **Zaid season** consistently presented significant economic challenges, often resulting in financial losses despite higher average water usage.

The Most Important Patterns Discovered

1. **Seasonal Profitability Disparity:** The stark contrast in profitability between Kharif (highly profitable), Rabi (moderately profitable), and Zaid (consistently unprofitable) was a primary pattern.
2. **Crop-Specific Seasonal Suitability:** Certain crops, like Sugarcane and Chilli, proved to be profitable across all seasons, suggesting their robustness or high market value, while staple crops (Wheat, Maize, Pulses, Rice) often struggled, particularly in Zaid.
3. **Environmental Influence:** Zaid season's lower rainfall and higher temperatures significantly impact water usage and yield, highlighting a strong environmental dependency.
4. **Irrigation Efficiency:** Drip irrigation consistently demonstrated superior profitability compared to other methods, suggesting its economic advantage.
5. **Yield-Profit Correlation:** A strong positive correlation between yield per hectare and profit was observed, underscoring the importance of maximizing output for economic gains.

The Most Useful Analytical Findings

- **Unprofitability of Zaid for Staples:** The consistent negative average profits for staple crops during the Zaid season is a crucial finding for risk management and crop selection.
- **Drip Irrigation's ROI:** The clear economic benefits of drip irrigation provide a data-backed argument for its promotion and adoption.
- **Kharif Season's Importance:** Its significant contribution to overall production and profitability makes it a focal point for agricultural policy and support.

- **Environmental Challenges in Zaid:** The characterization of Zaid by high temperatures and low rainfall, leading to high water usage and low profitability, is key for developing targeted interventions.

Important Limitations of the Analysis

- **Causation vs. Correlation:** While strong associations were found (e.g., between rainfall and yield, drip irrigation and profit), the analysis primarily indicates correlation rather than definitive causation.
- **Aggregated Averages:** Many findings are based on aggregated averages (e.g., average profit by season or irrigation method). This might mask profitable niches or specific local successes that deviate from the mean.
- **Data Granularity:** The dataset doesn't allow for fine-grained analysis of specific farming practices, detailed input costs, or precise market dynamics at a hyper-local level, which could influence individual farm profitability.
- **External Factors:** Factors such as government policies, pest/disease outbreaks (beyond the risk percentage), and global market prices were not deeply explored but could significantly impact performance.

How the Analysis Could Support Future Investigation or Decision-Making

This analysis provides a robust foundation for strategic decision-making and future research:

- **Informed Policy Making:** Agricultural bodies can use these findings to develop targeted support programs for Kharif, promote efficient irrigation methods, and devise risk mitigation strategies for the challenging Zaid season.
- **Farmer Guidance:** Farmers can receive data-driven advisories on optimal crop choices for each season, particularly for the Zaid season, and understand the potential benefits of adopting drip irrigation.
- **Resource Allocation:** Insights into water usage and profitability can guide more efficient allocation of resources (e.g., water, subsidies for irrigation technology) to maximize economic returns.
- **Further Research:** The identified patterns and correlations (e.g., between environmental factors and yield) can serve as hypotheses for more in-depth studies, potentially involving time-series analysis for seasonal trends, or experimental

designs to establish causality for irrigation methods and yield-enhancing practices.

- **Investment Decisions:** Investors in the agricultural sector can leverage these insights to identify high-potential crops and regions, or to fund sustainable farming technologies like drip irrigation.